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Paying Too Much? Borrower Sophistication and Overpayment in the U.S. Mortgage Market

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ABSTRACT

Comparing mortgage rates that borrowers obtain to rates that lenders could offer for the same loan, we find that many homeowners significantly overpay for their mortgage, with overpayment varying across borrower types and with market interest rates. Survey data reveal that borrowers' mortgage knowledge and shopping behavior strongly correlate with the rates they secure. We also document substantial variation in how expensive and profitable lenders are, without any evidence that expensive loans are associated with a better borrower experience. Despite many lenders operating in the U.S. mortgage market, limited borrower sophistication may provide lenders with market power.

SURVEY DATA INDICATE THAT HALF of the borrowers taking out a mortgage in the United States seriously considered only one lender, with just 3% considering more than three lenders. Ninety-five percent of respondents reported that they were satisfied that they received the lowest interest rate for which they could qualify.[1] Taking these facts at face value, one might conclude that there is little variation in mortgage pricing, or that borrowers are very efficient at finding the best rates. This conclusion might seem reasonable especially when considering that the mortgage market appears highly competitive: the

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[1] Statistics are based on the National Survey of Mortgage Originations (NSMO), which covers mortgage originations from 2013 to 2020.

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majority of mortgages in the United States are standardized and guaranteed by the government, and hundreds of lenders offer mortgages on any given day. However, in contrast to borrowers' perceptions, we find that many borrowers substantially overpay for their mortgage, and that overpayment varies systematically across borrower types and over time. We argue that this overpayment reflects, at least in part, limited borrower sophistication, which provides lenders with market power.

To assess overpayment, we use unique data on both available rates—the rates that lenders could offer for specific mortgages/borrowers in each market and day—and the mortgages locked, or obtained, by consumers. The available rates are inclusive of the fees and markups that a borrower would pay if they chose a particular lender. Importantly, these "offer" rates are lenders' private information rather than publicly posted, and may not be automatically offered to prospective borrowers. The data on locked mortgages include key variables for evaluating mortgage pricing, including several that are unavailable in any other data set, such as "discount points," exact time of the rate lock (as opposed to the closing date), and the lock period (e.g., 30 or 60 days).

For a given borrower, we compare the rate they locked against the distribution of available rates for the same type of loan and borrower (same loan-to-value [LTV] ratio, credit score, points, etc.) on the same day in the same market. Motivated by a simple search model (e.g., Carlson and McAfee (1983)), we construct a metric that we refer to as the *Expected Gain from Additional Search (EGain)*—the expected amount by which a borrower could reduce their rate if they were to obtain one additional rate quote.

We find that *EGain* differs substantially across borrower types. For example, borrowers getting Federal Housing Administration (FHA) loans—who tend to have lower incomes and credit scores—have an average *EGain* of 28 basis points (bp), and, remarkably, for a quarter of these borrowers *EGain* exceeds 40 bp. In contrast, borrowers getting "jumbo" loans—who tend to have high incomes—have an average *EGain* of just 4 bp, implying that such borrowers are much more likely to obtain rates that come close to the best available. We interpret these results as consistent with borrowers across different market segments varying in their financial "sophistication," which we think of as encompassing factors such as the recognition that lenders may offer different rates, and the ability to search and negotiate for low rates.

We also find that the expected gain from search varies over time, inversely with interest rates. When market interest rates (and lenders' offer rates) are higher, average *EGain* is lower, implying that borrowers tend to find better deals when rates are high. We argue that this relationship is driven in part by "behavioral" factors: when the level of rates is low, borrowers may feel less compelled to search for a good deal or negotiate hard than when rates are higher, even though in dollar terms the consequences are the same. Indeed, we also find explicit evidence from survey data on mortgage borrowers that shopping effort increases with market rates.

We provide additional evidence of ineffective shopping and negotiation when we analyze price dispersion in the locks data alone, where we have a much

larger sample and can further exploit unique features of the locks data. We find that the difference between the 90th and 10th percentile interest rate that observably identical borrowers lock in for the same loan in the same market on the same day with the same points is 55 bp; for a typical loan of $250,000, this corresponds to an upfront payment of about $6,250. Furthermore, the largest residual dispersion occurs for borrowers who may be the least financially sophisticated. For example, the 90–10 residual rate gap for FHA borrowers is about 72 bp.[2]

We then unpack the level at which this dispersion occurs—for example, across lenders versus within lenders, branches, or even loan officers. Estimated lender fixed effects exhibit substantial variation but still only reduce the residual rate dispersion by only a modest amount, suggesting limited explanatory power of factors such as (time-invariant) lender reputation or quality. Allowing lender fixed effects to vary by loan and borrower characteristics as well as over time improves explanatory power. However, even after further including branch and loan officer fixed effects, almost one-half of the residual dispersion remains, consistent with an important role for negotiation in this market.

In the final part of the paper, we present several findings that support the idea that mortgage lenders exert market power that allows them to charge markups, and that a lack of borrower financial sophistication is a source of lender market power. First, we document that the largest lenders tend to be the most expensive (i.e., have high residualized rates) and exhibit the most dispersion in their (residualized) rates, suggestive of large lenders having market power. Next, we use detailed income and expense data for mortgage lenders from regulatory filings merged with our data on transacted rates to show that relatively expensive lenders have higher profit margins (net income per dollar originated). We also find that expensive lenders spend slightly more on occupancy and technology, but substantially more on the salaries of loan officers and managers. This result could imply a better borrower experience, but our evidence from survey data indicates that borrowers do not receive better service when paying more for their mortgage. Finally, again using survey data, we show that the most sophisticated borrowers (those who shop the most and have high mortgage knowledge) obtain significantly cheaper mortgages than the least sophisticated borrowers. Moreover, this gap widens as local mortgage market concentration declines, suggesting that the benefit of more intense lender competition primarily accrues to sophisticated borrowers.

Overall, our empirical results indicate that a large fraction of the borrower population in the United States overpays for mortgages, and a lack of financial sophistication is a key reason. The borrowers who fare the worst often get government-guaranteed loans through the FHA program, which is aimed at lowering the cost of homeownership for lower-income households. Our results

[2] Notably, the lending platform providing our data is used mostly by monoline nonbank mortgage originators, and wide dispersion remains when we limit the sample to such lenders alone. We can therefore rule out cross-selling or bundling of other services as explanations for the observed patterns.

suggest that government entities such as the FHA might consider ways to reduce price dispersion and excessive markups to help fulfill their policy objectives. Our findings also suggest that the lack of consumer shopping is important for the pass-through of monetary policy to the mortgage market: reduced search effort appears to prevent borrowers' rates from falling as much as they could when market rates decrease, weakening the pass-through of expansive policy.

This paper makes several contributions to the literature on overpayment and dispersion in the price of mortgages and consumer credit more broadly. Previous work in the United States assesses dispersion in offer rates (Alexandrov and Koulayev (2017) and McManus et al. (2018)); transacted mortgage rates (Agarwal et al. (2024) and Ambokar and Samaee (2019)); reset rates on adjustable-rate mortgages (Gurun et al. (2016)); and mortgage broker fees (Woodward and Hall (2012)). Our data contain key details not available in any other data set (e.g., exact lock date, lock period, and points), which provides greater confidence that we are identifying true dispersion. Moreover, our data span multiple market segments, which allows us to show, for example, that dispersion is especially wide for FHA loans, and come from recent years after the implementation of postcrisis mortgage regulations (e.g., loan officer compensation rules) that could have reduced price dispersion. While we find considerable dispersion in mortgage rates, our estimates are substantively smaller than other estimates in the literature, as we discuss in a more detailed review of the literature in Section I of the Internet Appendix.[3] In addition, we document sizeable dispersion within lender, branch, and even loan officer, suggestive of a role for negotiation, in contrast to assumptions in previous work that negotiation plays little role in the U.S. mortgage market (Alexandrov and Koulayev (2017)).

Most notably, this paper is the first to compare transacted rates to offer rate distributions for identical loans, which allows us to measure the gains to additional search across all borrowers, including those at the bottom of the transaction rate distribution, and to examine how gains vary by borrower type. We also document that borrower overpayment and shopping intensity change with market rates over time, consistent with behavioral factors affecting shopping effort.[4]

[3] The Internet Appendix may be found in the online version of the article. Some work also exists outside the United States, where the institutional details and the mortgage market structure are different. Allen et al. (2014b) study the Canadian market, where there is no dispersion in posted rates but large dispersion in contracted rates, which they argue arises due to differences in bargaining leverage across consumers. Iscenko (2018) and Coen et al. (2023) document substantial heterogeneity in terms of how well U.K. borrowers fare when choosing mortgages from the options available at their lender; Liu (2019) shows that many U.K. borrowers appear to neglect nonsalient fees and that lenders exploit this in their price setting. Damen and Buyst (2017) provide evidence that mortgage borrowers in Belgium who shop more achieve substantial savings.

[4] The result that overpayment increases when market rates are low is distinct from and complements the finding of Fuster et al. (2024) that lender offers tend to feature higher markups when market rates are low.

Finally, this paper is the first to link data on firm-level mortgage pricing with revenues, costs, and profits. We confirm that our residualized lock rates are strongly correlated with firm revenues, and also show that expensive lenders earn higher margins per dollar originated. Alongside these results, we provide novel evidence that borrowers who pay more are less satisfied with their lender, and that a lack of borrower sophistication may be undermining competition in this market.[5]

Our paper connects to the literature on (in)efficiency of consumer choice and price dispersion in various consumer finance markets, including mutual funds (Hortaçsu and Syverson (2004), Choi et al. (2010)), auto loans (Argyle et al. (2022)), and credit cards (Stango and Zinman (2016)).[6] In addition to mortgages being the largest household liability, the composition of mortgage borrowers spans the income, wealth, and financial sophistication spectrums, providing much cross-sectional variation to help shed light on the factors driving dispersion and overpayment in consumer financial markets more broadly. Along with other work documenting costly "mistakes" in the mortgage market (e.g., Agarwal, Grigsby, and Yao (2015), Keys et al. (2016), Agarwal, Ben-David, and Yao (2017), Andersen et al. (2020)), our results are in line with the growing literature pointing at financial literacy/sophistication as a key driver of differential outcomes in household finance (e.g., Hastings et al. (2013), Gomes et al. (2021)).

In recent work, Agarwal et al. (2024) argue that overpayment by certain groups need not imply that they are unsophisticated (or have high search costs), but rather could be a rational response of relatively risky borrowers who fear being rejected. They document that the relationship between the contracted mortgage rate and the number of "inquiries" recorded by credit bureaus—their proxy for borrower search—is U-shaped. This suggests that borrowers who search a lot may do so because their application gets rejected, which may in turn lead these borrowers to accept relatively worse offers. This channel may contribute to some of the overpayment we document. At the same time, however, we find considerable overpayment even among many well-qualified borrowers, and we provide evidence from NSMO data that variation in sophistication is important to understand cross-sectional dispersion.[7]

[5] Our finding that lower market concentration may have limited benefits for less-sophisticated consumers is consistent with the evidence in Allen et al. (2014a) based on mergers in the Canadian mortgage market. Malliaris et al. (2022) also use NSMO data to document that indicators of sophistication are correlated with rates, and study how the benefits of knowledge and shopping interact.

[6] More broadly, a large literature studies price dispersion in various markets empirically and theoretically (see, e.g., Baye et al. (2006) and Wright et al. (2021) for surveys of relevant work). One important conclusion from this literature is that a decrease in information cost (making shopping easier) does not necessarily reduce equilibrium price dispersion—neither empirically nor theoretically.

[7] Over the period we study, underwriting standards in the Government-Sponsored Enterprise (GSE) and FHA segments of the market are largely dictated directly by these institutions. Thus, for the vast majority of borrowers who get approved for a loan, it should also be easy to get a loan

The rest of the paper is organized as follows. Section I provides background on the institutional details of this market. Section II describes our data sources. Section III explores how locked rates compare to the offer distribution; here we explain our *EGain* measure and study how it varies across borrowers with different characteristics and over time. Section IV assesses mortgage price dispersion in the locks data alone. Section V explores the connections between borrower sophistication, mortgage rates, and market power. Finally, Section VI concludes with potential policy implications.

I. Mortgage Pricing and Originations in the United States

In this section, we provide a brief overview of some of the institutional details that will be important for the rest of the paper.[8]

In the United States, there are multiple channels through which a borrower can obtain a loan. One is to go directly to a bank or credit union. An alternative is to obtain a loan through a specialized mortgage originator, a so-called "mortgage bank" or "independent mortgage company." Contrary to what the name suggests, these lenders are not depository institutions, and they typically do not keep any of the mortgages they originate on their own balance sheets. Finally, it is possible to go through a mortgage broker, who may have relationships with bank and nonbank originators and who acts as an intermediary connecting borrowers to those institutions. When a loan is originated directly by a lender who will either retain the loan on portfolio or sell it in the secondary (mortgage-backed security, or MBS) market, it is called a "retail loan"; if a loan is originated via a nonbank entity that originates the loan for another lender, it is called "wholesale loan."

Regardless of the channel, a borrower generally interacts (in person or by phone/online) with a loan officer or broker (henceforth "LO") who will have access to various "rate sheets" that provide the detailed pricing available at a given point in time (generally updated at least once a day). Importantly, for any loan type and combination of characteristics, there is no single interest rate—instead, the rate sheet shows a combination of note rates and "(discount) points." To obtain a low note rate, a borrower can pay points (where 1 point = 1% of the loan amount). If the borrower is willing to take a higher rate, they can receive points (often called rebates or credits) that can be used toward the origination costs.

In the case of a retail loan, the available pricing will come directly from the lender's pricing desk; in the case of wholesale lending, the rate sheets can come from several different lenders (often referred to as "investors"). Each rate sheet will provide pricing for different loan programs (e.g., GSE loans, FHA, or jumbos) with adjustments depending on a few loan and borrower characteristics,

from a different lender. That said, the *perception* that other lenders are unlikely to accept one's application may suffice to induce a borrower to accept a relatively "bad" offer.

[8] For additional discussion, see, for example, Fuster et al. (2013) or https://files.consumerfinance.gov/f/201301_cfpb_final-rule_loan-originator-compensation.pdf.

typically FICO credit score, LTV, loan amount, geographic region, loan purpose, and property type. Pricing depends on the value that a lender assigns to the loan—often based on the current value of such a loan in the MBS market, where most loans are ultimately sold.[9] Prices also take into account required "guarantee fees" charged by the GSEs and Ginnie Mae.[10] Furthermore, lenders will add a margin that may depend, among other things, on the level of demand for loans (Fuster et al., 2024).

On top of the prices from the rate sheet, the costs to the borrower include compensation of the LO and/or their employer (e.g., the mortgage bank). This compensation may be explicit (via upfront origination fees) or implicit (via lender profit margins on rate sheets). Historically, LOs had strong incentives to sell loans with higher interest rates, all else equal, and thereby generate more compensation not only for the lender but also for themselves (often called the "yield spread premium"; see Woodward and Hall (2012)). In the wake of the financial crisis, new regulations were imposed so that LO compensation may no longer vary with the interest rate and other terms of the loan. Lenders still profit, of course, when borrowers take higher interest rates.[11] Importantly, however, this does not imply that all LOs in a firm simply get paid an identical, fixed amount for each loan they originate. Indeed, LOs are frequently given a choice between different possible compensation plans, for example, trading off fixed salary for higher commission rates per dollar of originated loans.

Finally, it is not the case that rate sheets together with a specific LO's compensation plan always determine the final rate and points/fees that a given borrower is offered—"exceptions" may be granted, for instance, to meet a competitive outside offer. Lenders generally have specific procedures for these exceptions to avoid violating fair lending laws.[12]

An important step in the origination process is the mortgage rate lock. A lock is a guarantee that the borrower will be issued a mortgage with a specific combination of interest rate and points if the mortgage closes by a specific date. Borrowers typically lock their mortgage rates as protection against rate increases between the time of the lock and the time when the mortgage closes (e.g., for 30 or 60 days). A lock can occur when a borrower submits a loan

[9] Generally, prices in the MBS market depend on the yields on alternative investments (especially Treasuries) as well as investors' projections of future prepayments of the underlying mortgages.

[10] In addition to the guarantee fee, which is a flow insurance premium over the life of a mortgage, the GSEs charge upfront "loan-level price adjustments" that depend on borrower and loan characteristics—see, for example, https://www.fanniemae.com/content/pricing/llpa-matrix.pdf. As for Ginnie Mae, it charges a fee to guarantee the timely payment of principal and interest to investors in MBS backed by loans that are insured by the FHA or guaranteed by the Department of Veterans Affairs (VA).

[11] These rules were first changed in 2011 as part of the Truth in Lending Act; the Consumer Financial Protection Bureau (CFPB) published its final rule on LO compensation requirements in January 2013.

[12] See, for example, https://www.crai.com/sites/default/files/publications/Managing-the-Fair-Lending-Risk-of%2DPricing-Discretion-Whitepaper-Oct-2014.pdf or https://www.mortech.com/mortechblog/pricing-discretion-fair-lending-risk.

application with a lender, but can also occur at a later time. Not all rate locks ultimately lead to originated mortgages, since the loan application can still be rejected afterward (e.g., because the home appraisal comes in lower than expected) or the borrower could renege. However, the lock is binding on the lender as long as the characteristics of the loan and borrower (e.g., the loan amount or credit score) remain as specified at the time of the lock. Lenders typically do not charge an explicit fee for a rate lock, although loan application fees generally apply, and a longer lock period is often associated with a higher interest rate. Also, if a loan does not close by the time the lock period expires, extending the lock typically requires a fee.

II. Data

In this section, we describe our primary data sources: (i) data on mortgage rate locks and lender offers from Optimal Blue; (ii) data on nonbank income statements from Mortgage Call Reports (MCR) collected by state regulators; and (iii) survey data from the NSMO.

A. Optimal Blue Data

Our main data come from an industry platform called Optimal Blue that connects over 600 mortgage lenders with more than 200 whole loan investors. Through the platform, mortgage originators can gather information on mortgage pricing, initiate rate locks, manage pipeline risk, and sell mortgages to investors. Over 40,000 unique users access the system each month to search loan programs and lock in consumer mortgages. More than 2.4 million mortgage locks were processed through this system in 2019, accounting for about 30% of loan originations nationally.

The lenders using the platform tend to be nonbank monoline mortgage lenders. These lenders have gained substantial market share in the postcrisis period (see, e.g., Buchak et al. (2018); in 2019, they originated 56% of all purchase loans and 58% of refinance loans CFPB (2020)). Optimal Blue is also used by smaller community banks or credit unions. That said, many institutions on this platform act as correspondent lenders, meaning that they originate loans intended to be sold to other financial institutions such as a large bank like JP Morgan or Wells Fargo.

For this study, we use two components of the data generated by the platform: (i) data on mortgage products and mortgage prices actually accepted by consumers, and (ii) data on mortgage products available and mortgage prices offered by lenders.

A.1. Mortgage Rate Lock Data

The first source of data is the universe of "rate lock" agreements for the mortgages processed through the Optimal Blue platform. We have access to all the mortgage locks generated by the platform since late 2013. Since the market

coverage increases over the course of 2013 to 2014, we start using the data from January 2015; we end in December 2019. The data have wide geographical coverage of about 280 metropolitan areas as well as rural areas. All of the standard loan characteristics used for underwriting are included: LTV ratio, FICO score, debt-to-income (DTI) ratio, loan amount, loan program, loan purpose (purchase or refinancing), asset documentation, income documentation, employment status, occupancy status, property type, ZIP code location, etc.

These data have a number of unique features relative to servicing data that are typically used in mortgage research. First, the data include not only the contracted mortgage rate, but also the discount points or credits associated with that rate (meaning additional upfront payments made or received by the borrower). Second, we observe the exact timestamp of when the rate was locked, while in other data sets only the closing date is recorded, which generally differs from the pricing-relevant lock date by several weeks or even months. Finally, we have unique identifiers for the lender, branch, and the LO that processes each mortgage. For some lenders, we can also observe LO compensation, expressed as a percentage of the loan amount.[13]

While the lock data feature numeric lender identifiers, they do not directly provide information on the lenders. However, we are able to distinguish whether a subset of lenders are independent nonbanks or traditional depository institutions by relying on a match between Optimal Blue locks and Home Mortgage Disclosure Act (HMDA) data.[14] This will be useful later to assess whether lender type and cross-selling might be driving the patterns in the data that we document.

We restrict the sample in various ways to ensure that we study a relatively uniform set of loans that is representative of the type of mortgages originated in recent years. We only keep 30-year fixed-rate mortgages on owner-occupied single-unit properties with full documentation of assets and income, and we drop self-employed borrowers. We also drop loans for amounts under $100,000 and those with implausible values for LTV, DTI, or points/credits. Finally, we drop VA loans and streamline refinances (which comprise a small part of the sample). This leaves us with 3.6 million observations. For the analysis in Section III, we further restrict the sample to match the locked mortgages to offers for identical characteristics, as we describe below.

Table I presents summary statistics from the lock data sample that we use for the analysis in this paper, distinguishing between the four loan programs in the data, since they differ substantially in terms of borrower and loan characteristics. The four programs are: conforming (loans typically securitized through the GSEs Fannie Mae or Freddie Mac), super-conforming (with loan

[13] Some lenders process compensation outside of the Optimal Blue system, or do not compensate LOs directly on a per-loan basis.

[14] We use a loan-level merge between Optimal Blue, administrative FHA data, and HMDA used in Bhutta and Hizmo (2021), which covers the years 2014 to 2015. We also match Optimal Blue directly to the expanded public HMDA data from 2018 to 2019 to further classify lender types for lenders entering the Optimal Blue system after 2015.

Table I

Summary Statistics for Rate Lock Data

This table presents summary statistics for the Optimal Blue rate lock data. Our sample includes 30-year fixed-rate mortgages on owner-occupied single-unit properties, with full documentation of assets and income. Self-employed borrowers, loans for amounts under $100,000, VA loans, and streamline refinances are excluded.

	Conforming		Super-Conforming		Jumbo		FHA	
	Mean	St. Dev.	Mean	St. Dev.	Mean	St. Dev.	Mean	St. Dev.
Loan Amount (000)	255	94	544	71	720	262	222	92
Interest Rate	4.33	0.51	4.31	0.47	4.21	0.50	4.30	0.61
Discount Points Paid	0.15	0.95	0.28	0.97	0.19	0.74	0.06	1.14
FICO	742	47	750	41	763	33	669	47
LTV	81	14	80	12	77	10	93	8
DTI	35	9	36	9	31	9	42	10
First-Time Homebuyer %	24		23		11		49	
Refinance Share %	31		33		33		17	
Observations	2,316,400		119,894		76,941		1,092,535	

amounts above the national conforming limit but below the local limit; the GSEs can still securitize these loans, but potentially at slightly worse prices), jumbo (loan amount above the local conforming limit, meaning the loan cannot be securitized through the GSEs), and FHA loans (which require mortgage insurance from the FHA and are typically pooled into securities guaranteed by the government entity Ginnie Mae).

The table shows that FHA loans are most likely to go to first-time homebuyers with low FICO scores and high LTVs and DTIs. Jumbo loans, the only loan type for which the credit risk is not guaranteed by the government, tend to go to the most creditworthy borrowers and feature relatively low LTVs. These loans constitute only about 2% of our sample. The table also shows that FHA borrowers on average pay fewer discount points than borrowers in the other programs; Figure IA.1 in the Internet Appendix displays the cumulative distribution of points paid (or received) by program.

As noted above, not all lenders use the Optimal Blue platform, and not all rate locks necessarily result in an originated mortgage. Thus, there is a concern that the distribution of interest rates recorded in our rate lock data may not accurately represent the rates that borrowers ultimately end up with. However, in Section II of the Internet Appendix, we show that the interest rates observed in the rate lock data mirror the interest rates observed in the well-known McDash mortgage servicing data set on originated mortgage loans, in terms of both averages and dispersion. Furthermore, loan/borrower characteristics in Optimal Blue locks also look very similar to those in data on originated loans.[15]

A.2. Mortgage Offers Data

As our second source of data, we collect information on the menu of mortgage products and mortgage rates that lenders offer through the platform's pricing engine. Optimal Blue's "Pricing Insight" allows users (e.g., LOs) to retrieve the real-time distribution of offers for a loan with certain characteristics in a given local market (where an offer consists of a combination of a note rate and upfront fees and points that the borrower pays or receives with this rate). Importantly, the offers we observe are "customer facing," that is, rates are inclusive of margins and fees that borrowers could expect to pay at a particular lender. The Insight interface is designed for lenders to compare their pricing against that of peers.[16]

[15] See Table IA.II and Figure IA.4 in the Internet Appendix for details. For jumbo mortgages, the locked interest rates in Optimal Blue tend to be higher than those in McDash, which could indicate that the relatively smaller lenders who use the Optimal Blue platform may not be as competitive for these types of loans as for FHA and conforming loans. It is also the case that average jumbo loan amounts are somewhat smaller in Optimal Blue locks than in McDash originations, which could reflect some differential selection of borrowers. The dispersion of rates remains very similar, however.

[16] The Pricing Insight data are different from earlier Optimal Blue data used by Fuster et al. (2024). Those data were based on rate sheets that did not include LO compensation and origination charges, unlike the Pricing Insight data that we use here (where offers are "all included").

For any combination of day, MSA, and loan/borrower characteristics, we measure an "offer" rate for each lender on the platform. This offer rate reflects the interest rate (with zero points) that the lender could offer a prospective borrower, including fees under the assumption that the loan is originated by the LO that has locked the most loans for that lender in that market.[17]

If a lender represents multiple different investors, the offer we observe is based on the most competitive investor offer. As a result, a borrower locking a loan with this lender will not necessarily get the exact observed offer rate, for three reasons. First, the locked rate can vary depending on which LO the borrower goes through, since different LOs may charge different markups. Second, the LO may offer a loan that is not based on the rate sheet of the most competitive investor, but rather on one from a different investor.[18] Third, as noted earlier, borrowers may be able to negotiate and get an "exception" or a lower rate from the lender.

We conduct daily searches in one local market (Los Angeles), twice-weekly searches in four markets, and weekly searches for 15 additional markets.[19] We collect offer distributions for 100 loan types that differ across the following dimensions: FICO score, LTV ratio, loan program, loan purpose (purchase or cash-out refinance), occupancy (owner-occupied or investor), rate type (30-year fixed or 5/1 adjustable), and loan amount. The mortgages require full documentation of income, assets, and employment, and are secured by single-unit homes.

An important limitation of the offers data is that we are not able to track institutions over time or match them directly to the lenders in the lock data, since there is no fixed lender identifier. The time series is also slightly shorter than for the locks data, as we started systematically tracking offers in April 2016.

Figure 1 shows the dispersion in mortgage rates available from different lenders, pooling data over time and across all of the 20 metropolitan areas for which we obtained data. To make offers comparable, we subtract the median offer rate across lenders for the same product, day, and metropolitan area. Figure 1 indicates wide dispersion in offer rates. In particular, we find a 53 bp difference between the 10th and 90th percentile offers, similar to what Alexandrov and Koulayev (2017) and McManus et al. (2018) document. In Section III of the Internet Appendix, we further show that the degree of offer rate dispersion is quite similar across types of loans and borrowers, and across all 20 cities in our sample.

[17] As we explain further in Section III of the Internet Appendix, we observe a distribution of prices (points) for a given note rate, which we transform into a distribution of rates for zero points.

[18] One reason an LO might want to do this is to maintain active relationships with multiple investors.

[19] The markets with twice-weekly searches are New York City, Chicago, Denver, and Miami. The markets with weekly searches are Atlanta, Boston, Charlotte, Cleveland, Dallas, Detroit, Las Vegas, Minneapolis, Phoenix, Portland, San Diego, San Francisco, Seattle, Tampa, and Washington DC.

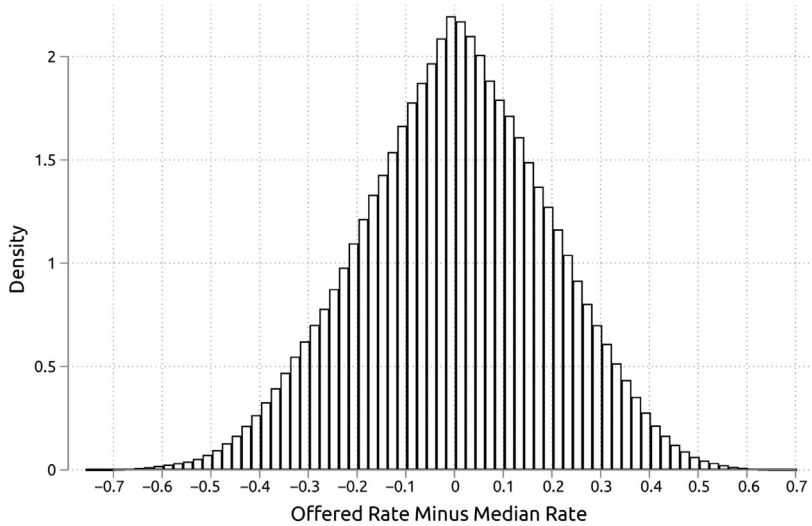

Figure 1. Offer dispersion for identical mortgages. This figure shows the distribution of real-time offered interest rates, in percentage points, where for each offer rate we subtract the median offered rate across lenders for an identical mortgage in the same metropolitan area. The histogram includes data between April 2016 and December 2019 from 20 metropolitan areas for 52 combinations of loan characteristics (FICO, LTV, program, loan amount). Data Source: Optimal Blue.

B. MCR Data

We use firm-level data on mortgage-origination-related income and expenses of nonbank mortgage lenders from MCR.[20] Nonbank financial institutions ("shadow banks") that hold a state license through the Nationwide Multistate Licensing System (NMLS) must regularly complete an MCR, which contains data on mortgage loan activity (e.g., total mortgage origination volume) and on firm financial conditions, including detailed information on income and expenses related to mortgage lending.

To study how variation in mortgage pricing across firms relates to firm income, expenses, and profitability, we merge the MCR data with the Optimal Blue locks data. The MCR data cover about 90% of nonbank mortgage originations in 2018 to 2019; our MCR-Optimal Blue matched data cover 197 firms operating as independent nonbank mortgage lenders in 2018 or 2019 that accounted for about 40% of nonbank mortgage originations in 2018 and 2019. This lower coverage in our matched data set relative to the MCR data mainly reflects the fact that not all nonbank lenders use the Optimal Blue platform.[21]

[20] These data are available to Federal Reserve researchers through an agreement with the Conference of State Bank Supervisors, which owns and operates the system that collects MCR data on behalf of state regulators. In addition to income and expense data, firms report data on assets and liabilities. Similar data are used by Jiang et al. (2020) to study the capital structure of nonbanks.

[21] These coverage ratios use HMDA data to measure total market size (see Bhutta et al. (2017) for more on HMDA data coverage and nonbank market share). We merge the MCR and Optimal

C. NSMO Data

The NSMO is a joint initiative of FHFA and CFPB as part of the National Mortgage Database program. The NSMO surveys a nationally representative sample of borrowers with newly originated closed-end first-lien residential mortgages, focusing in particular on borrowers' experiences getting a mortgage, their perceptions and knowledge of the mortgage market, and their future expectations. We use data from the first 26 waves, including borrowers who took out mortgages from 2013 through 2019.[22] In addition to its unique focus on new mortgage borrowers, a key feature of the NSMO is that survey respondents are matched with extensive administrative data from mortgage servicing records and consumer credit files. We therefore observe precise data on the credit risk of respondents (e.g., their credit score, DTI ratio, etc.) and the terms and conditions of the mortgages that respondents obtained.

For our analyses in this paper, we impose several sample restrictions. We only consider mortgages on a household's primary residence and drop mobile and manufactured homes as well as two- to four-unit dwellings. In addition, we focus on fixed-rate loans with a term of 30 years, and we drop construction loans or loans obtained through a builder, mortgages with an associated additional lien, and mortgages with more than two borrowers on the loan. Finally, we drop a few observations for which the survey respondent was not a borrower on the loan. These restrictions leave us with 22,567 mortgages for the analysis.[23] In all of our NSMO analyses, we use the provided analysis weights, which are based on sampling weights and nonresponse adjustments.

III. Comparing Locked Rates to Offer Rates

In this section, we study whether different types of borrowers get good or bad deals, relative to what is available in the market at the time they lock their mortgage. This analysis allows us to assess which types of borrowers tend to "overpay" for their loans, and which factors drive differences across borrowers and over time.

To do so, we use the distribution of lenders' offer rates (described in Section II.A.2) by day, MSA, FICO, LTV, loan amount, and loan program (i.e., conforming, super-conforming, jumbo, and FHA). We match locked loans to the offers available for loans with nearly identical characteristics—see Section IV of the Internet Appendix for additional details.[24] We focus the analysis on

Blue data sets on lender name, after first mapping Optimal Blue lender IDs to lender names by conducting a loan-level match to HMDA data. We match about 450 mortgage lenders operating in 2018 or 2019 from Optimal Blue to HMDA, of which about 240 are independent nonbanks.

[22] The public version of the data are available on FHFA's website. We are able to use a confidential version of the data that includes geographic information.

[23] The full NSMO data set contains 39,615 loans originated before 2020. Limiting attention to 30-year mortgages drops just over 10,000 observations; most of the non-30-year mortgages have a term of 15 years. These sample sizes are as of July 2023, but the data may be revised in the future.

[24] The offer rates are for a loan with zero points and fees. To compare to these offers, we adjust the locked rate for points paid or received by the borrower based on the empirical relationship

purchase mortgages from the 20 MSAs for which offer data are collected, and end up with 67,537 matched loans.

To compare the locked rate to the offers, the main metric we consider is the *Expected Gain from Additional Search (EGain)*. This metric is motivated by a simple search model (e.g., Carlson and McAfee (1983)).[25] Suppose n mortgage lenders are posting mortgage rate offers for a particular borrower type. Rates are ordered from lowest to highest: $r_1 \leq r_2 \leq \ldots \leq r_n$. Borrowers only see the mortgage rates available at the lenders they meet with, and we assume that each borrower has an equal chance of meeting any one of the lenders. Suppose a borrower has already found a rate r_k and is considering whether to engage in search to obtain one additional rate quote from a new lender. Given that the probability of meeting a lender (different from lender k) that offers rate r_i is $1/(n-1)$, the expected gain from additional search is given by

$$EGain_k = \sum_{i=1}^{k-1}(r_k - r_i)\frac{1}{n-1} = \left[r_k - \sum_{i=1}^{k-1}\frac{r_i}{k-1}\right]\frac{k-1}{n-1}. \tag{1}$$

Intuitively, the term in brackets is the locked rate minus the expected rate available at the $k-1$ lenders offering rates lower than r_k. Of course, the borrower does not know which lenders are offering rates lower than r_k, so we have to adjust the expectation by the share of these lenders in the remaining population, which is $(k-1)/(n-1)$. This measure therefore gives us how much money the borrower is leaving on the table, in expectation, by not conducting one more search. The measure takes into account not just where the borrower's rate is relative to the center of the distribution, but also the width of the offer distribution—the expected gain from search is higher the more widely dispersed the offers are. Furthermore, in our implementation we account for the fact that some lenders do not offer a given loan type at all, which would correspond to $r_i = \infty$.[26] However, it is important to note that the measure takes the distribution of offers at a point in time as given, while, if all borrowers were to search more, the equilibrium distribution of offers might shift. It is therefore best to consider it as a measure of overpayment from the perspective of a single borrower.

We also consider an alternative simpler measure where the locked rate is compared directly to the median offer rate for matched loans—we call the difference between the two the *Locked-Offer Rate Gap*. Positive values indicate that a borrower overpays relative to what a typical lender (as measured by the median) could offer for the same loan characteristics on the same day.

One might be concerned that the distribution of offers could be a flawed benchmark for locked rates if the best offers are not "achievable" for some

between points and interest rates. See Section V of the Internet Appendix for more details on how we estimate this empirical relationship.

[25] We are grateful to Gregor Matvos for suggesting this measure to us.

[26] We therefore take n to be the number of unique lenders in a given MSA on a given day that offer at least one loan type across all programs. However, setting n equal to the number of lenders who post offers for a given loan type does not qualitatively change the results.

reason. However, we note that overall, 5.3% of all borrowers obtain a rate in the best (bottom) 5% of their offer distribution. Even for FHA borrowers, which we find below to do relatively poorly on average compared to the available rates, this fraction is 3.5%. Thus, even the best offers are indeed available to borrowers.[27]

A. Cross-Sectional Differences in Expected Gain from Additional Search

Panel A of Figure 2 shows the distribution of the expected gain from search for all mortgages in our data. The dashed vertical line denotes the mean of the distribution. The variable *EGain* is by definition bounded below at zero. The figure shows that about one-quarter of locked loans would in expectation gain less than 5 bp from an additional search. However, the distribution is highly skewed, and there are borrowers who could expect to lower their rate by 50 bp or more by randomly contacting another lender for an offer on the exact same loan type.

Panel B of Figure 2 and the summary statistics in Table II show that the distributions of *EGain* vary substantially across loan programs. In particular, *EGain* is largest for FHA borrowers, at 28 bp on average. Moreover, one-quarter of FHA borrowers would in expectation gain 40 bp or more from an additional search. In contrast, the distributions for super-conforming and jumbo borrowers are very different: even at the 75th percentile, the expected gain from an additional search is only 12bp for super-conforming loans and 3 bp for jumbo loans, indicating that many borrowers in these segments come close to the best offers posted on Optimal Blue.

Table II further shows that borrowers with lower FICOs and higher LTVs typically have higher remaining gains from search and overpay the most relative to the median available offer. Importantly, this overpayment is relative to offered rates that already incorporate risk-based pricing for low FICOs and high LTVs. First-time homebuyers and those that pay discount points also tend to fare worse.[28] Finally, borrowers at independent nonbanks appear to overpay by more, on average, than borrowers at other lender types; we return to potential differences across lender types below.

It is worth noting that within each of the groups considered in Table II, there is substantial dispersion in *EGain* and the locked-offer rate gap, as shown by the usually large gaps between the 75th and 25th percentiles. Thus, even for high-FICO or low-LTV borrowers, a nontrivial fraction of borrowers lock rates well above what other lenders could offer them. However, dispersion tends to be largest for the groups that on average fare the worst.

[27] In addition, in Figure IA.3 in the Internet Appendix, we confirm that offer rates derived from Optimal Blue Insights closely track offer rates for comparable loans published by Mortgage News Daily, an industry website, as well as by Zillow and Freddie Mac's Primary Mortgage Market Survey.

[28] Note that since we adjusted the mortgage note rate for points paid, this relationship is not "mechanical."

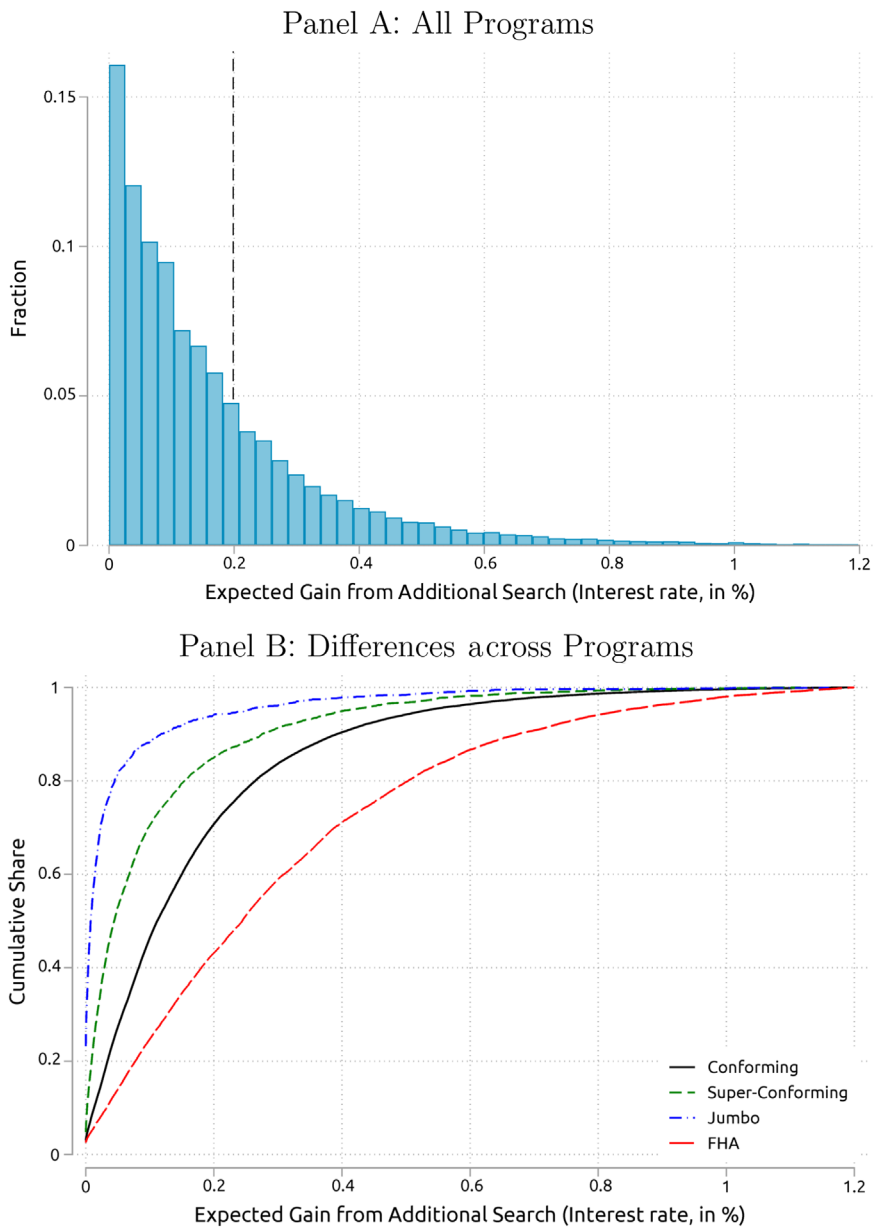

Figure 2. Distribution of the expected gain from additional search for identical mortgages. Panel A shows the distribution of the expected gain from additional search, defined in equation (1), for all borrowers in the sample. Being at (or close to) zero means that the rate locked by a borrower is near the lowest available rate, so that the borrower would have little to gain from additional search. Larger values mean that a borrower would have more to gain from additional search. The dashed line denotes the mean of the distribution. Panel B shows the cumulative share of loans below a certain expected gain across the four loan types in the sample. (Color figure can be viewed at wileyonlinelibrary.com)

Table II

Summary Statistics for Expected Gain from Search and Locked-Offer Rate Gap

This table presents summary statistics for the expected gain from an additional search, which is given by equation (1), and of the locked-offer rate gap, which is the difference between each locked rate and the median offer rate in the same market on the same day for an identical mortgage. In the discount points category, negative values mean that the borrower receives points (also known as a rebate or credit) while positive values mean that the borrower pays points.

		Expected Gain from Search			Locked-Offer Rate Gap		
	Observations	Mean	25th pct.	75th pct.	Mean	25th pct.	75th pct.
All Mortgages	67,637	0.18	0.04	0.25	0.10	−0.08	0.25
Program							
FHA	14,715	0.28	0.08	0.40	0.25	0.01	0.46
Conforming	46,535	0.17	0.04	0.22	0.09	−0.06	0.22
Super-Conforming	4,448	0.10	0.01	0.12	−0.05	−0.21	0.08
Jumbo	1,939	0.04	0.00	0.03	−0.23	−0.35	−0.08
FICO							
[640, 660)	7,629	0.27	0.06	0.40	0.23	−0.02	0.46
[680, 700)	9,617	0.22	0.05	0.31	0.16	−0.06	0.35
[720, 740)	10,666	0.19	0.04	0.26	0.11	−0.06	0.26
740+	39,725	0.15	0.04	0.21	0.07	−0.09	0.20
LTV							
(75, 80]	22,270	0.13	0.03	0.17	0.02	−0.12	0.16
(85, 90]	7,295	0.15	0.04	0.21	0.05	−0.09	0.20
(90, 95]	16,573	0.16	0.04	0.22	0.08	−0.08	0.21
(95, 97]	21,499	0.27	0.08	0.38	0.23	0.01	0.42
First-Time Homebuyer							
No	33,378	0.15	0.03	0.20	0.06	−0.09	0.20
Yes	34,253	0.21	0.05	0.30	0.15	−0.05	0.31
Discount Points							
[−5, −0.2)	14,133	0.14	0.02	0.19	0.00	−0.17	0.19
[−0.2, 0.2]	23,430	0.16	0.03	0.21	0.08	−0.10	0.21
(0.2, 5]	30,074	0.22	0.06	0.30	0.18	−0.01	0.31
Lender Type							
Independent Nonbank	56,483	0.19	0.05	0.26	0.12	−0.05	0.27
Other/Unclassified	11,154	0.13	0.02	0.17	0.00	−0.15	0.16

Table IA.XII in the Internet Appendix provides analogous summary statistics based on the median income, college education share, minority share, and mortgage market concentration in a borrower's location.[29] Average *EGain* is largest in areas with low household income, low shares of college-educated residents, and high minority shares; these are also the areas in which the dispersion in this measure is larger. In contrast, there is little covariation between the *EGain* distribution and local mortgage market concentration.

[29] Income, education, and minority shares are measured at the ZIP-code level based on 2017 American Community Survey data; mortgage market concentration is the Herfindahl-Hirschman Index (HHI) at the county level averaged over 2016 to 2019 using HMDA data.

Table III

Regressions of the Expected Gain from Search on Observables

This table presents regressions of the expected gain from an additional search, given by equation (1), on bins of borrower FICO score or mortgage LTV and on loan officer compensation. The data cover mortgage rates for 20 metropolitan areas during the 2016 to 2019 period. We focus on 30-year, fixed-rate, fully documented purchase mortgages. Standard errors shown in parentheses are two-way clustered at the month and lender level. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	(1)	(2)	(3)	(4)	(5)	(6)
FICO (omitted cat.: [640,660)):						
$I_{680 \leq FICO < 700}$	-0.039^{***}	-0.029^{***}	-0.024^{***}			
	(0.006)	(0.005)	(0.008)			
$I_{720 \leq FICO < 740}$	-0.069^{***}	-0.047^{***}	-0.045^{***}			
	(0.008)	(0.006)	(0.009)			
$I_{FICO \geq 740}$	-0.097^{***}	-0.067^{***}	-0.056^{***}			
	(0.008)	(0.006)	(0.008)			
LTV (omitted cat.: (60,80]):						
$I_{85 < LTV \leq 90}$				0.015^{***}	0.011^{***}	0.011^{***}
				(0.003)	(0.003)	(0.004)
$I_{90 < LTV \leq 95}$				0.033^{***}	0.025^{***}	0.021^{***}
				(0.004)	(0.004)	(0.004)
$I_{LTV > 95}$				0.128^{***}	0.102^{***}	0.074^{***}
				(0.009)	(0.008)	(0.011)
Loan Officer Comp (%)			0.107^{***}			0.096^{***}
			(0.025)			(0.027)
Loan Amount F.E. ($10k bins)	Yes	Yes	Yes	Yes	Yes	Yes
MSA × Month F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Lender-Branch F.E.		Yes	Yes		Yes	Yes
Adj. R^2	0.122	0.403	0.403	0.160	0.429	0.417
Observations	67,637	65,757	15,444	67,637	65,757	15,444

A.1. Regression Analysis

The summary statistics above indicate that borrowers who are financially less well-off, such as lower-FICO and higher-LTV borrowers, overpay the most. As noted above, this overpayment is relative to offered rates that already incorporate risk-based pricing for low FICOs and high LTVs. Instead of reflecting risk-based pricing, overpayment could be driven by borrowers who are less financially sophisticated, meaning that they are less effective at searching for and negotiating low rates. However, an alternative potential explanation for these patterns unrelated to sophistication is that lower-FICO borrowers and higher-LTV borrowers tend to have smaller loans and thus less of an incentive (in dollar terms) to shop around.

In columns (1) and (4) of Table III, we regress *EGain* on bins for different FICO scores and LTV ratios, respectively, as well as fine loan amount bins and MSA-by-month fixed effects. Conditional on loan amount, we find that lower-FICO borrowers and higher-LTV borrowers continue to overpay more, to a similar degree as we observe in Table II. Thus, such borrowers appear to obtain more expensive loans (relative to available offers) for reasons beyond the differential incentive to shop stemming from loan size variation.

Another potential explanation for why low-FICO and high-LTV borrowers are more likely to pay too much is that they sort into more expensive lenders. Borrowers might choose expensive lenders because those lenders offer better service or because they spend more on marketing and are more visible. To investigate this explanation, we add lender-branch fixed effects in columns (2) and (5) of Table III. In these columns, the R^2 jumps sharply to about 40% from less than 20%, meaning that lender-specific pricing differences explain a fair amount of variation in *EGain*. Furthermore, the coefficients on FICO and LTV become slightly smaller, implying that sorting into lenders does explain some of the overpayment by low-FICO and high-LTV borrowers. Still, even within the same branch, these borrowers tend to pay differentially more relative to what is available in the market. This result suggests that they may be less effective at negotiating a good rate.

Along similar lines, low-FICO and high-LTV borrowers may sort into LOs that charge more because these LOs may be more experienced and offer more assistance in the loan application process. To test for this possibility, in columns (3) and (6) we control directly for LO compensation, which typically ranges from 1% to 2% of the loan amount, for the subset of lenders who report it. The coefficients on this variable are strongly significant; however, the coefficients on FICO and LTV change little, implying that low-FICO and high-LTV borrowers do not pay higher rates simply because they match with expensive LOs. Rather, these borrowers appear to pay more than other borrowers even when working with LOs with the same compensation—again suggesting that these borrowers may be less effective in shopping for and negotiating a good rate.

Robustness. Table IA.XIII in the Internet Appendix reproduces these regressions with the locked-offer rate gap as the dependent variable; the patterns are similar. In another robustness check, reported in Table IA.XIV, we restrict the sample to lenders who we can identify as independent nonbanks. Doing so leaves the coefficients from Table III essentially unchanged. As noted earlier, the nonbank lenders who constitute the majority of our sample are only in the business of originating mortgages. Thus, differences in *EGain* cannot be explained by potential price advantages that bank lenders might grant to financially well-off (high-FICO, low-LTV) customers, for instance, because they also have significant account balances or other business with the bank.[30]

Finally, it may be the case that many of the lenders making offers in our data set are small and difficult for consumers to find. To ensure our results are unaffected by this possibility, we replicate our analysis using only offers from high-volume lenders, as designated on the Optimal Blue platform. The results, reported in Table IA.XV in the Internet Appendix, again remain qualitatively unchanged.

[30] Table II showed that borrowers who obtain loans fron nonbanks tend to have higher *EGain* and locked-offer gaps; this could reflect either overall advantageous pricing by banks/credit unions or selection by borrowers. What we emphasize here is that such differential pricing, if it exists, does not appear to vary with borrower creditworthiness.

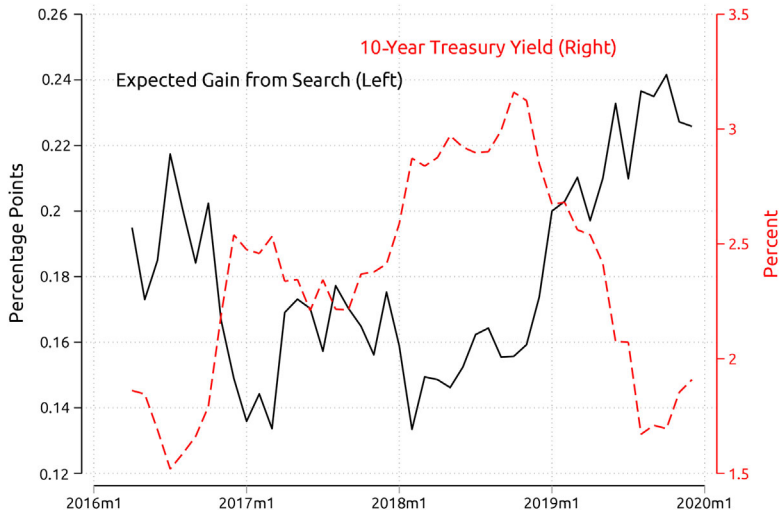

Figure 3. The evolution of average expected gain from additional search and Treasury yields. The dashed red line in the figure is the 10-year Treasury yield. The solid black line is the monthly average expected gain from additional search (*EGain*) after controlling for borrower and loan characteristics (FICO, LTV, loan amount, and MSA fixed effects). (Color figure can be viewed at wileyonlinelibrary.com)

B. Time-Series Movements in Expected Gain from Additional Search

The last section explored the cross-sectional patterns in the expected gain from additional search. In this section, we instead study how *EGain* moves over time, with a particular focus on how it responds to changes in market interest rates. Are borrowers more likely to end up with worse rates (relative to what is offered in the market) when market rates are low, and more likely to get a good deal as rates increase? If so, what might explain this relationship?

Figure 3 plots the average *EGain* against market interest rates, here measured by the 10-year Treasury yield.[31] In the summer of 2016, the level of market interest rates as reflected by Treasury yields was very low. At the same time, *EGain* was high, meaning that borrowers were locking rates from the higher end of the offer rate distribution. As Treasury yields increased, and as a result lenders increased their offer rates, *EGain* shrunk, indicating that borrowers moved toward the cheaper end of the offer distribution. When rates fell again starting in late 2018, the inverse happened. Overall, the movements in average *EGain* almost mirror the movements in Treasury yields.

[31] For the average *EGain* series, we use the estimated month fixed effects from a regression similar to those in Table III but controlling simultaneously for FICO and LTV. We use the 10-year Treasury yield as our measure of market rates since it is strongly correlated with the 30-year fixed mortgage rate, but avoids potential endogeneity issues due to the measurement of the latter. However, using the mortgage rate or the current-coupon MBS yield instead leaves our conclusions unchanged.

Table IV

The Relationship between the Expected Gain from Search and Treasury Yields

This table presents regressions of the expected gain from an additional search, given by equation (1), on Treasury yields and other controls. *Treasury Yield* is the daily 10-year yield on the day of the mortgage rate lock, and $DTI \leq 36$ is a dummy variable equal to one if a borrower's debt-to-income ratio does not exceed 36%. All specifications include controls for FICO, LTV, and loan amount, as well as for ZIP code house price growth (from Zillow) over the past 12 months. The data cover mortgage rates for 20 metropolitan areas during the 2016 to 2019 period. We focus on 30-year, fixed-rate, fully documented purchase mortgages. Standard errors shown in parentheses are two-way clustered at the month and lender level. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	(1)	(2)	(3)	(4)	(5)	(6)
Treasury Yield	−0.051***	−0.044**	−0.026**	−0.057***	−0.051***	−0.031**
	(0.007)	(0.018)	(0.012)	(0.007)	(0.018)	(0.012)
Treasury Yield × DTI ≤ 36				0.013**	0.015***	0.012***
				(0.005)	(0.005)	(0.004)
Borrower and Loan Controls	Yes	Yes	Yes	Yes	Yes	Yes
MSA F.E	Yes	Yes	Yes	Yes	Yes	Yes
MSA × Month F.E.		Yes	Yes		Yes	Yes
Lender-Branch F.E.			Yes			Yes
Adj. R^2	0.145	0.163	0.430	0.146	0.165	0.430
Observations	67,241	67,241	65,349	67,241	67,241	65,349

We confirm the statistical significance of the relationship between *EGain* and market rates in Table IV.[32] The first column regresses *EGain* on the 10-year Treasury yield, controlling for all borrower characteristics jointly, MSA fixed effects, and house price growth at the ZIP code level over the past 12 months to account for housing market "hotness," which could affect borrowers' willingness to spend time shopping around. The coefficient implies that as the 10-year Treasury yield increases by one percentage point, the average *EGain* falls by about 5 bp. This effect is sizable given that over our sample as a whole, *EGain* averages 18 bp.

In column (2), we add month fixed effects (interacted with MSA) and see that the relationship between Treasury yields and *EGain* is similarly strong within-month. Column (3) further adds lender-branch fixed effects, to examine the extent to which the estimated relationship gets weaker once we control for potentially time-varying selection of borrowers into expensive or cheap lenders/branches. The coefficient on the Treasury yield declines (to 2.6 bp), suggesting that some of the overall relationship may be due to borrowers selecting cheaper lenders when rates are higher, consistent with additional shopping.

In the remaining three columns, we test the extent to which this relationship may be driven by affordability constraints: when market rates rise, constrained borrowers may be forced to shop more to find a relatively good rate that will allow them to qualify for a mortgage (Bhutta and Ringo (2021)). To test this mechanism, we interact the Treasury yield with an indicator

[32] Table IA.XVI repeats the analysis with locked-offer rate gap as the dependent variable.

for borrowers with a DTI no higher than 36% (borrowers who are likely unconstrained by the payment burden from higher rates).[33] The interaction term in columns (4) to (6) is positive and significant, meaning that the relationship between *EGain* and market rates is indeed weaker for unconstrained borrowers. However, the magnitude of the estimated coefficients is relatively small, meaning that even for unconstrained borrowers, *EGain* is higher when market rates are lower.

The results above suggest that the relationship may be driven at least in part by "behavioral" factors: when the level of rates is already low, borrowers may feel less compelled to search for a good deal or negotiate hard than when rates are higher, even though in dollar terms the consequences are the same. This might be the case particularly after a recent drop in rates, as borrowers might compare their offer to a higher reference level.

In Section VII.B of the Internet Appendix, we use the NSMO data to provide direct evidence that shopping effort increases with market rates. For example, we find that when market rates are higher, recent mortgage borrowers are more likely to report having considered or applied to more than one lender. These patterns hold even after controlling for local house price growth to account for variation in market "hotness," which might be correlated with interest rates and affect borrowers' ability to spend time shopping for a cheaper mortgage.

Importantly, the higher *EGain* when market interest rates are low is in addition to the higher "price of intermediation" when rates are low, as identified by Fuster et al. (2024). That paper shows that *offers* feature higher lender markups (relative to loan values in the MBS market) at times of high demand, and that this pattern is driven at least in part by lender capacity constraints.[34] Thus, there are two complementary reasons why after a drop in market rates, borrowers obtain worse mortgage rates than they could in a frictionless world: lenders make worse offers relative to an MBS-market benchmark, and borrowers fare worse relative to those offers.

IV. Understanding Variation in Locked Rates

In the previous section, we observed that many borrowers overpay relative to the rates available for the same mortgage in the same location on the same day. Furthermore, this overpayment is amplified when market interest rates are low. In this section, we take a more comprehensive look at the dispersion in rates locked in by borrowers who are identical on observables (including the day of the lock and the location). This analysis is closer to existing literature on this topic, but the additional detail available in our locks data allows us to

[33] Using alternative DTI cutoffs to separate borrowers (e.g., 43%) leaves the results qualitatively unchanged.

[34] Capacity constraints could also affect relative bargaining power and contribute to the patterns documented in this section. When market rates are low and mortgage demand is very high, a borrower may have less bargaining power because an LO may not care as much about losing a customer, since many others are waiting to refinance and the LO is already at or close to capacity.

more precisely measure and unpack price dispersion than what earlier work has been able to do. See Section I of the Internet Appendix for a detailed review of the literature and how our analysis is distinct from it.

To investigate dispersion in locked mortgage rates, we regress locked rates on borrower and loan characteristics, as well as time effects, and then add an increasingly fine set of fixed effects. Our outcome of interest is the remaining dispersion in the residual, which we measure in terms of standard deviations and as the gap between 75th and 25th or 90th and 10th percentiles. Comparing the residual dispersion along with the adjusted R^2 across specifications allows us to assess the relative importance of different drivers of price dispersion in this market.

Table V shows the results from various specifications, estimated on the same set of nearly 3 million loans locked over the 2015 to 2019 period.[35] In the first column, as a benchmark, we include only lock date-by-MSA fixed effects, to document the amount of overall interest rate dispersion within the same MSA on the same day. These day-by-MSA fixed effects explain just under 60% of the total variation in rates, and the standard deviation of the residual is 33 bp. Given the importance of pure time-variation to explain variation in rates, in the remaining columns we also report an adjusted R^2 within day and MSA (i.e., after absorbing the first set of fixed effects).

In column (2), we add our baseline set of controls, an extensive set of underwriting variables that consist of fully interacted bins of values for FICO, LTV, and loan program, interacted with lock month to allow for time-variation in risk pricing.[36] We also include borrower ZIP code fixed effects, lock period fixed effects, property type fixed effects, cubic functions of loan amount and DTI, and linear controls for FICO and LTV (to allow for within-bin variation).[37] This specification is similar to regressions one could typically run with a mortgage servicing data set.[38] Within day and MSA, the adjusted R^2 from these variables is 0.36, and substantial dispersion remains: the standard deviation in residuals is 0.26, and the borrower at the 90th percentile of the residual distribution pays 58 bp more than the borrower at the 10th percentile.

[35] The estimation drops "singleton" observations that are determined entirely by the set of fixed effects. There are more such singletons as we add more fixed effects. To ensure that our results are not driven by changing samples, we use the sample from the most restrictive specification (10) in all specifications. Using the largest possible sample for each specification instead does not materially affect the results.

[36] We include 13 FICO bins, 9 LTV bins, and 12 dummies for the four loan programs interacted with three loan purposes (purchase, rate refinance, and cash-out refinance). The choice of FICO and LTV bins is motivated by the loan-level price adjustments set by the GSEs.

[37] The lock period typically varies from 15 to 90 days, with 30 and 45 days being the most common choices. A longer lock period leads to a slight increase in the fee (or equivalently, the interest rate).

[38] It is already somewhat more precise, since here we control for the date when a loan is locked, along with the length of the lock period, while in typical data sets loans originated in a given month may have been locked in different months. In Section I of the Internet Appendix, we report an alternative and more realistic comparison.

Table V

Unpacking the Dispersion in Locked Interest Rates

This table presents regressions studying the amount of variation in locked mortgage rates explained by different sets of explanatory variables. The dependent variable is the mortgage interest rate locked. The data cover mortgage rates locked for 277 metropolitan areas during the 2015 to 2019 period. We focus on 30-year, fixed-rate, fully documented mortgages. "Program" refers to 12 dummy variables representing four loan programs interacted with three loan purposes. Specifications (2) to (10) also include lock period fixed effects, property type fixed effects, cubic functions of loan amount and DTI, as well as linear functions of FICO, LTV, and (specification (3) onward) discount points. For MSAs that span multiple states, we include MSA × state fixed effects.

	Underwriting Grid			Add Lender Controls			Add Branch Controls		Add LO Controls	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Standard Deviation	0.33	0.26	0.24	0.22	0.21	0.18	0.16	0.15	0.15	0.13
75–25 Percentile	0.36	0.28	0.26	0.22	0.21	0.17	0.15	0.14	0.13	0.12
90–10 Percentile	0.79	0.58	0.55	0.48	0.44	0.38	0.33	0.31	0.30	0.26
Underwriting Variables Grid										
Lock Date × MSA F.E.	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FICO × LTV × Program × Lock Month F.E.		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
ZIP Code F.E.		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Discount Points × Program × Lock Month F.E.			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Add Lender Controls										
Lender F.E.				Yes	Yes	Yes	Yes	Yes	Yes	Yes
Lender × Lock Date F.E.					Yes	Yes	Yes	Yes	Yes	Yes
Lender × FICO × LTV × Program × Lock Month F.E.						Yes	Yes	Yes	Yes	Yes
Lender × Points × Lock Month F.E.						Yes	Yes	Yes	Yes	Yes
Add Branch Controls										
Branch F.E.							Yes	Yes	Yes	Yes
Branch × Lock Month F.E.								Yes	Yes	Yes
Add Loan Officer Controls										
Loan Officer F.E.									Yes	Yes
Loan Officer × Program F.E.										Yes
Loan Officer × Lock Year F.E.										Yes
Adj. R^2	0.58	0.75	0.77	0.81	0.82	0.85	0.88	0.88	0.89	0.90
Adj. R^2 (Within Date × MSA)		0.36	0.42	0.52	0.55	0.62	0.69	0.70	0.72	0.74
Observations	2,996,149	2,996,149	2,996,149	2,996,149	2,996,149	2,996,149	2,996,149	2,996,149	2,996,149	2,996,149

Column (3) adds bins for the points paid or received by the borrower (interacted with program by lock month).[39] This (usually unobserved) variable does indeed explain some of the rate differences across borrowers, but substantial dispersion remains—for example, the 90th to 10th percentile difference is still 55 bp. The adjusted R^2 of 0.42 indicates that standard underwriting variables and upfront payments explain less than half of the variation in interest rates paid across borrowers within the same day and MSA.

Based on the regression coefficient on discount points (not shown in the table), we can translate interest rates to upfront points. This coefficient implies that one discount point changes the interest rate by about 21.8 bp on average (see Section V of the Internet Appendix for details). Therefore, a rate difference of 55 bp is approximately equivalent to 2.5 upfront discount points or 2.5% of the mortgage balance. In other words, our results imply that a borrower with a $250k mortgage who borrows at the 90th percentile interest rate should be getting—but in fact is not getting—lender credit of about $6,250 relative to someone borrowing at the 10th percentile interest rate. Alternatively, if one prefers to think in terms of mortgage payments, 55 bp corresponds to about $80/month for a $250k loan at the average level of rates over our sample period.

Table VI shows how the residual dispersion in interest rates varies across loan programs and characteristics. The middle column of the table uses the residuals from specification (3). We see an extreme amount of dispersion for the two lowest FICO groups. We also see substantial dispersion for FHA-insured loans, despite the fact that these loans are fully insured by the government and thus lenders and investors take very little, if any, credit risk. In other words, it is highly unlikely that unobserved risk factors could explain the wide dispersion in FHA interest rates. Similarly, we find fairly wide dispersion for conforming and super-conforming loans, which meet the credit standards of the GSEs and are likely purchased and fully guaranteed by these institutions.[40] Finally, we continue to see wide dispersion when we focus just on low-risk borrowers, that is, those with prime FICO scores in excess of 680 and those with LTVs of less than 75%.

Jumping back to Table V, in column (4), we add lender fixed effects to allow for the possibility that some of the rate dispersion may reflect differences in lender characteristics such as service quality or advertising costs; we study correlates of these estimated fixed effects in Section V. We find that adding

[39] We include eight point bins, as well as a linear function in points to allow for within-bin variation.

[40] One caveat here is that lenders may be worried about so-called "put-back" risk, where loans in default must be repurchased by the lender due to some defect in the underwriting found by the FHA or GSEs. In the case of the GSEs, Goodman (2017) documents that put-back risk has been negligible since lenders have stopped issuing low-documentation and other nontraditional loans. For FHA loans, perhaps the biggest concern for lenders has been litigation risk under the False Claims Act, which allows the federal government to sue lenders who knowingly submit false or fraudulent claims to the FHA. Under the Obama Administration, some of the largest lenders settled with the government, paying fines close to $5 billion. That said, this risk is most salient for large banks with significant capital at risk, unlike the nonbanks that dominate our data. Furthermore, this risk has eased in recent years.

Table VI

Summary Statistics for Residualized Locked Rate

This table shows the gap between 90th and 10th percentiles in the residualized locked mortgage rate from specifications (3) and (10) of Table V.

		90th to 10th Percentile Gap	
	Observations	Spec. (3) of Table V	Spec. (10) of Table V
All Mortgages	2,996,149	0.55	0.26
Program			
FHA	884,681	0.72	0.31
Conforming	2,001,083	0.48	0.25
Super-Conforming	72,419	0.43	0.21
Jumbo	37,966	0.48	0.24
FICO			
<600	42,369	0.93	0.47
[600, 640)	221,431	0.82	0.37
[640, 680)	490,343	0.68	0.30
[680, 740)	933,110	0.55	0.27
≥740	1,308,896	0.44	0.23
LTV			
≤75	506,400	0.47	0.23
(75, 80]	624,315	0.46	0.23
(80, 95]	968,492	0.50	0.26
>95	846,179	0.71	0.33
First-Time Homebuyer			
No	1,991,677	0.50	0.24
Yes	1,004,130	0.64	0.31
Loan Purpose			
Purchase	2,294,776	0.55	0.27
Cashout	351,269	0.50	0.23
Rate Refi	350,104	0.57	0.25
Lender Type			
Independent Nonbank	1,897,661	0.55	0.27
Other/Unclassified	1,098,488	0.54	0.26

these lender effects decreases the 90th to 10th percentile difference by 7 bp and increases the within-day-and-MSA adjusted R^2 by 0.1. In columns (5) and (6), we additionally interact the lender fixed effects with lock-day fixed effects and other controls, to allow for the possibility that lenders' (relative) pricing may change over time or may differ across loan types. This reduces the 90–10 gap by a further 10 bp from column (4), and adds another 0.1 to the adjusted R^2. These results suggest that in addition to time-invariant differences across lenders, price dispersion may reflect lender pricing strategies that vary over time and across programs. Such variation would make it difficult for borrowers to find low rates simply by following the recommendations of family, friends, or real estate agents—yet this is a common approach borrowers take to finding a mortgage.

In columns (7) and (8), we further allow pricing to vary across different branches of a lender. As discussed earlier, the lenders in our data set tend

to be nonbank monoline mortgage lenders and community banks. For a typical lender in our data, in a given MSA most loans are originated through just two or three branches located within that MSA. Differential branch pricing could reflect differences in convenience of the office location and/or costs (e.g., office rent). In addition, as noted earlier, different branches can have different markups and pricing strategies.[41]

The branch fixed effects in column (7) have noticeable incremental explanatory power, increasing the adjusted R^2 and reducing the residual dispersion. Adding branch-by-month fixed effects in column (8) has more modest effects. Column (8) should come close to looking at observably identical borrowers getting a loan from the same branch at the same time, yet the 90–10 gap remains at 31 bp and the interquartile range at 14 bp.

Lastly, in columns (9) and (10), we further allow pricing to vary across LOs in the same branch, which could reflect, for instance, differences across LOs in terms of experience, compensation, or willingness/ability to negotiate. Which LO a borrower matches up with (within a branch) does appear to matter somewhat for the rate they end up with, since the adjusted R^2 further increases and the residual dispersion decreases in the last two columns. Nevertheless, even after including LO fixed effects that are allowed to vary across both time and programs, the 90–10 gap remains at 26 bp.

The last column of Table VI shows that the cross-sectional patterns in residual dispersion, already discussed above, remain similar in the most restrictive specification (10): the dispersion is substantially larger for FHA loans, low-FICO borrowers, or first-time homebuyers. The final rows of the table show that the residual dispersion is identical if we only consider loans that were locked with lenders who we are able to classify as independent nonbanks (as discussed in Section II.A.1). This suggests that the large dispersion is not driven by unobservable pricing adjustments that banks or credit unions might make for customers who already have accounts or other business with them.

To summarize the findings from this analysis, there is a large amount of dispersion in the rates that observably identical mortgage borrowers pay, even after controlling for the exact timing and upfront payments. Adding lender, branch, and LO controls reduces the residual rate dispersion by about half. Substantial dispersion remains, however, implying that two observably identical borrowers may get quite different deals from the same lender branch or even the same LO at the same time. Moreover, this pattern appears to be more pronounced for financially less well-off and potentially less sophisticated borrowers.

V. Borrower Sophistication and Lender Market Power

Thus far, we have presented evidence indicating that many borrowers lock in mortgage rates that are high relative to available offers in the market. We

[41] We discuss correlates of estimated branch fixed effects (within lender) in Section VI of the Internet Appendix. See also the discussion in Section V.A below.

have also seen that there is considerable dispersion in locked rates both within lender and within location (e.g., ZIP code). These results suggest that price differences across borrowers cannot be fully explained by lender-specific attributes (e.g., service quality) or local market area characteristics (e.g., lender concentration). Moreover, we find that rate dispersion and the expected gains from additional search are highest among borrowers who may be the least financially sophisticated, such as low-FICO and FHA borrowers. Taken together, these facts suggest an imperfect market in which lenders are able to charge markups to many borrowers, especially those who may lack the ability to shop and negotiate effectively.

As noted earlier, in most U.S. cities there are many active lenders, and market concentration is low (Amel et al. (2018)). With many active firms selling a highly standardized product, it may be surprising to observe significant markups and price dispersion. Theoretically, however, less concentration may not always improve competition and reduce markups. For example, in the model of Armstrong and Vickers (2022), where consumers only consider subsets of firms when choosing from which one to buy, entry does not always lower prices (see also (Gabaix et al. (2016)). As Syverson (2019) notes, firms have "market power" when they can influence the price they charge as they do not face a perfectly elastic demand curve, and concentration can be a poor proxy for market power.

In this section, we use additional data sources to provide further evidence that price dispersion reflects an imperfect market in which lenders exert market power, as borrowers lack the ability to shop effectively. This section proceeds in four subsections. In the first, we measure lender "expensiveness" and document which firm characteristics (e.g., size) are correlated with expensiveness. In the second, we use the matched MCR-Optimal Blue data to test whether expensive lenders earn higher profits or have higher costs. In the last two subsections, we turn to the NSMO data. We use these data to assess whether more expensive lenders differentiate themselves by providing better service to borrowers. We then test how borrowers' sophistication relates to the mortgage rates they get, and whether competition may be undermined by a lack of borrower sophistication.

A. Which Lenders are the Most Expensive?

We begin by studying whether variation in lender "expensiveness" is related to other lender attributes such as size and firm type (i.e., bank or nonbank). We measure expensiveness as the estimated lender fixed effects from specification (4) of Table V. There are 678 unique lenders in the data, but for this analysis we restrict attention to the 452 lenders who locked at least 100 loans in the sample used for the regressions in Table V. Figure 4 displays the distribution of fixed effects. The standard deviation of these fixed effects is about 0.19, or 19 bp. Thus, there is considerable variation across lenders in their average mortgage rate expensiveness, in addition to considerable *within*-lender dispersion as we showed earlier.

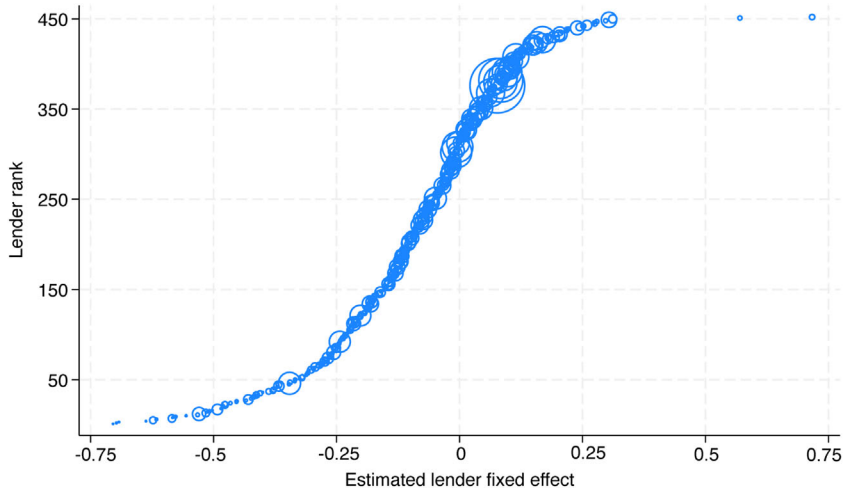

Figure 4. Distribution of estimated lender fixed effects from interest rate regressions. The figure shows estimated lender fixed effects, in percentage points, based on specification (4) in Table V. Only lenders with at least 100 locked loans in Optimal Blue are included in the figure; size of circles is proportional to total number of locks. (Color figure can be viewed at wileyonlinelibrary.com)

We next run lender-level regressions of the lender fixed effects on lender size, proxied here by the number of locks a lender makes in total, and lender type (nonbanks vs. banks). To facilitate interpretation, we simply divide lenders into size quartiles. Column (1) of Table VII shows that larger lenders tend to be significantly more expensive, and nonbanks are more expensive than banks in the sample (for some lenders, type cannot be determined, but the fixed effects for these lenders are not significantly different from those of the banks, which are the omitted category). Column (2) adds the share of locks by a lender that are FHA, jumbo, and super-conforming loans (with conforming loans as the omitted category). Lenders with higher shares of FHA loans are significantly more expensive, and adding this information on loan composition increases the explanatory power of the regression substantially.[42]

In the third column, we find that lenders with higher within-lender rate dispersion also tend to be more expensive.[43] Furthermore, column (4) shows that once we control for the lender size, loan type, and variation in the residual, FHA share and within-lender dispersion remain highly statistically significant

[42] Recall that the lender fixed effects are estimated from regressions in which loan type, as well as many other variables, is controlled for. As a result, this regression does not reflect that FHA loans are more expensive; rather, it indicates that lenders who make many FHA loans are on average more expensive across all loan types.

[43] Dispersion is measured as the standard deviation of residuals from specification (6) of Table V, that is, after accounting for lender fixed effects that are further allowed to vary across loan types and over time. The 10th percentile of this variable is 0.07, and the 90th percentile is 0.20.

Table VII

Which Lenders Are the Most Expensive?

This table displays lender-level regression results using data for 452 lenders who use the Optimal Blue platform and have at least 100 mortgage locks. The outcome variable in columns (1) to (4) are the lender fixed effects from specification (4) in Table V. Within-lender dispersion is measured as the standard deviation of residuals from specification (6) of Table V. Robust standard errors are in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Dependent Variable:	Lender FE	Lender FE	Lender FE	Lender FE	Within-Lender Dispersion Dispersion
	(1)	(2)	(3)	(4)	(5)
Size Quartile 2	0.0533**	0.0488*		0.0253	0.0216***
	(0.0258)	(0.0256)		(0.0238)	(0.00672)
Size Quartile 3	0.0740***	0.0812***		0.0282	0.0487***
	(0.0256)	(0.0243)		(0.0234)	(0.00577)
Size Quartile 4	0.127***	0.131***		0.0465*	0.0777***
	(0.0244)	(0.0235)		(0.0257)	(0.00573)
Nonbank	0.108***	0.0622***		0.0579***	0.00395
	(0.0220)	(0.0215)		(0.0202)	(0.00499)
Lender Type Unknown	0.00916	0.0151		−0.0147	0.0273***
	(0.0215)	(0.0210)		(0.0208)	(0.00550)
FHA Share		0.385***		0.267***	0.108***
		(0.0763)		(0.0767)	(0.0169)
Jumbo Share		−0.154		−0.253**	0.0910
		(0.202)		(0.105)	(0.110)
Super-Conforming Share		0.0405		0.0460	−0.00502
		(0.214)		(0.206)	(0.0449)
Within-Lender Dispersion			1.637***	1.091***	
			(0.177)	(0.203)	
Constant	−0.205***	−0.296***	−0.317***	−0.355***	0.0540***
	(0.0234)	(0.0266)	(0.0251)	(0.0267)	(0.00687)
Adj. R^2	0.16	0.28	0.22	0.34	0.37
Observations	452	452	452	452	452

while the effect of lender size is reduced. This is due in part to larger lenders exhibiting more variation in the rate residual (see the last column of Table VII).

In sum, we find that larger lenders and those that originate a larger share of FHA mortgages are more expensive on average, and also exhibit more variation in the mortgage rates their borrowers lock, after finely accounting for observable characteristics. One possible interpretation of these findings is that larger lenders may have market power that enables them to charge more on average, while still offering competitive rates to some borrowers (likely those that shop around and/or negotiate).[44] Alternatively, some lenders may be able to have high market share and charge higher prices if they can differentiate

[44] The Armstrong and Vickers (2022) model predicts that larger firms choose higher maximum prices, in line with our evidence. In their model (where firms set a range of prices and play mixed strategies), this is because some consumers only consider one firm they are aware of and so large firms have less to gain from setting lower prices to attract consumers who consider multiple firms.

themselves by providing borrowers better service. Sections V.B and V.C below look at whether loan pricing is related to lender costs and customer satisfaction.

A.1. Branch Expensiveness

In Section VI of the Internet Appendix, we describe a similar analysis to understand pricing variation across branches *within* lenders. Similar to the finding for lender expensiveness, we document that branches that originate more FHA loans are more expensive. We also find that the local characteristics of the areas in which branches are originating loans correlate with their pricing. For example, branches in areas with a lower share of college-educated population are more expensive. This finding is reminiscent of findings in Drechsler et al. (2017) that deposit spreads vary with local college-education shares, which suggests that a lack of depositor sophistication may be a source of market power in deposit markets. In Section V.D, we use the NSMO data to measure borrower sophistication more directly and at the individual level, and find that sophistication strongly predicts the rates borrowers get.

B. Do Expensive Lenders Earn Higher Profits?

In this subsection, we ask whether the large differences in loan expensiveness across lenders documented above are reflected in higher costs or in higher profitability reported by these lenders. To do so, we merge our measure of lender expensiveness from Optimal Blue with quarterly financial filings of these lenders (the MCR data; see Section II.B). For our sample period of 2015:Q1 to 2019:Q4, there are 1,897 lender-quarter filings from 162 unique lenders.

The first three columns of Table VIII provide summary statistics on the main line items from lenders' financial statements.[45] The median reported gross income is $4.75 per $100 originated, with most coming from secondary market income (or "gain-on-sale"), which is the income lenders earn from selling the loans they originate in the secondary market. The median of gross expenses for lenders is $4.16 per $100 originated, with the majority of expenses going toward personnel expenses. The second to last line shows after-tax net income for residential origination only, before corporate allocations. The median of this measure is $0.49 per $100 originated. The last line of the table shows net income across all lines of business (including servicing) after taxes and corporate allocations, with a median of $0.32 per $100 originated. Note that there is also substantial heterogeneity in all of these variables across lenders within the

[45] Both income and expenses correspond only to origination, warehousing, and secondary marketing of mortgages for one- to four-unit residential, and do not include income and expenses from other lines of business such as servicing, multifamily/commercial, or residential property portfolio management.

Table VIII

Nonbank Income and Expenses, and Their Relationship with Estimated Lender Expensiveness

This table is based on 1897 quarterly Mortgage Call Report filings from 162 unique lenders between 2015:Q1 and 2019:Q4. All variables are shown as dollars per $100 originated. Some categories for income and expenses are combined or not shown since they only have zeros for all firms. The standard deviation reported in the third column is the cross-sectional standard deviation, computed after taking out year-quarter fixed effects. The reported coefficients in the last two columns are from separate median regressions of each line item on lender expensiveness (in percentage points) as measured from lender fixed effects (from specification (4) of Table V) and year-quarter fixed effects. The standard errors determining statistical significance are clustered at the lender level. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

				Regression on Lender Expensiveness	
	Mean	Median	St. Deviation	β	St. Err.
Income					
Origination Income	1.33	0.80	1.53	0.78***	(0.19)
Interest Income	0.28	0.23	0.32	0.03	(0.04)
Secondary Market Income (Gain-on-Sale)	3.35	3.52	1.76	3.37***	(0.43)
Other Income	0.35	0.02	2.11	0.05**	(0.02)
Gross Income	5.03	4.75	1.97	4.05***	(0.41)
Expenses					
Loan Production Officers (Sales Employees)	1.37	1.36	0.58	1.06**	(0.47)
Loan Origination (Fulfillment/Nonsales)	0.65	0.58	0.42	0.45***	(0.13)
Origination-Related Management and Directors	0.28	0.15	0.36	0.25***	(0.07)
Employee Benefits	0.27	0.25	0.19	0.33***	(0.05)
Other Personnel Expenses	0.44	0.31	0.54	0.24*	(0.13)
Interest Expenses	0.25	0.21	0.37	0.06	(0.04)
Occupancy and Equipment	0.25	0.23	0.15	0.30***	(0.03)
Technology	0.10	0.08	0.09	0.12***	(0.03)
Outsourcing, Professional, and Subservicing Fees	0.15	0.09	0.22	0.10***	(0.02)
Other Noninterest Expenses	0.65	0.51	0.64	0.26	(0.23)
Gross Expenses	4.33	4.16	1.76	3.50***	(0.41)
Corporate Overhead					
Total Corporate Expenses	0.35	0.11	1.22	0.28***	(0.10)
Net Income					
Net Income (residential originations, precorporate allocations)	0.67	0.49	1.21	0.45**	(0.18)
Net Income (all lines, after corporate allocations and taxes)	0.31	0.32	1.87	0.24**	(0.12)

same quarter. The cross-sectional standard deviation of gross and net income is $1.97 and $1.87 per $100 originated, respectively.

We next investigate whether variation in lender "expensiveness" is related to these measures of lender income and costs. As in the previous subsection, we measure expensiveness as the lender fixed effects from specification (4)

of Table V. The last column of Table VIII displays results from separate median regressions of each line item on our measure of lender expensiveness (expressed in percentage points) and year-quarter fixed effects.[46] A one-percentage-point increase in interest rates charged by lenders is associated with extra gross income of $4.05 per $100 originated. The size of the effect is in line with expectations, as it implies that the value of one percentage point of interest rates to lenders is about four upfront points, which implies a reasonable point-rate trade-off of about 25 bp. Of the different income line items, lender expensiveness has the largest effect on secondary market income, although it also has a moderate effect on origination income (i.e., fees). While these results are not surprising, they confirm that our measure of lender expensiveness is related to income exactly as we would expect.

Turning to lender costs, a one-percentage-point increase in interest rates charged is associated with $3.5 of extra gross expenses per $100 originated. Most of these extra expenses reflect increased personnel expenses, particularly payments to LOs and managers. More expensive lenders do spend a bit more on technology, occupancy, and equipment but these effects are fairly small.

Lastly, we find a positive and significant effect on net income: a one-percentage-point increase in interest rates charged is associated with $0.45 of extra net income for $100 of residential loans originated, and $0.24 of extra net income across all lines of business after taxes and corporate allocations.

Overall, we find that the substantial variation in our estimated fixed effects (shown in Figure 4) is also reflected in the gross and net income that nonbank lenders report. Thus, it is not the case that the variation in interest rates across borrowers "nets out" within a lender such that some borrowers would cross-subsidize others but without an overall impact on the lenders; rather, some lenders are indeed more expensive and profitable than others.[47] Although more expensive lenders also have higher costs, these higher costs are not sufficient to offset their higher income. Perhaps even more importantly, the higher costs mostly reflect payments to LOs and managers; only to a minor extent do more expensive lenders spend more on technology or occupancy (e.g., office rental) that would arguably improve borrowers' experience. Of course, it is still possible that the more highly paid employees at expensive lenders provide borrowers superior service; we test this hypothesis in the next subsection.

[46] We focus on median regressions to minimize the effect of outliers and possible data errors in the filings data. In Internet Appendix Table IA.XVII, we report results from OLS (mean) regressions that are qualitatively similar, although less precise. That table also shows that additionally controlling for lender size has little effect on the estimated relationships between different line items and expensiveness.

[47] This point is further illustrated in Figure IA.2, which shows that the shares of borrowers who get particularly good deals or particularly bad deals are negatively related within lenders—not positively as in a world with pure cross-subsidization.

C. Are Expensive Lenders Better?

Above we find that lenders who charge more tend to have higher personnel costs. This leaves open the possibility that the LOs at expensive lenders are of "better quality" and provide better customer service, which in turn might justify their higher costs.

We cannot directly test this hypothesis within our Optimal Blue and MCR data, but we can do so using the NSMO survey data. This survey asks respondents a battery of questions about the quality of their experience in obtaining a mortgage. For example, the survey asks "yes/no" questions about whether borrowers experienced delays in their closing date and in paperwork processing. It also asks borrowers about their level of satisfaction ("very," "somewhat," or "not at all" satisfied) with their lender, with various aspects of the lending process, and with the interest rate that they got.

To test whether borrowers who pay more receive better service, we run a series of regressions of lender service-quality indicators on the interest rate borrowers obtained, conditional on a detailed set of controls:

$$Y_{ijtw} = \beta Rate_i + \Gamma Z_{ij} + \alpha_t + \delta_w + \epsilon_{ijtw.} \tag{2}$$

In equation (2), Y is the service-quality outcome for borrower i with mortgage characteristics j who originated a loan in month t and was surveyed in wave w, $Rate_i$ is the contract rate on the mortgage for borrower i, and Z_{ij} is a rich set of borrower and mortgage characteristics. The full list of controls is provided in the note to Table IX; it contains, for instance, flexible controls for credit score and LTV, fixed effects for county, program (e.g., GSE or FHA), and purpose (purchase or refinance), as well as borrower characteristics capturing income, employment, wealth, race, and more.[48] We further include origination month fixed effects α_t, which will absorb any economy-wide changes in service quality (e.g., when lenders hit capacity constraints), and survey wave fixed effects δ_w.

We emphasize here that most of the right-hand-side variables in equation (2) are drawn from administrative data, not from self-reported sources. Because of the large number of precisely measured loan and borrower controls, we interpret β as the relationship between service quality and mortgage expensiveness, as the controls account for variation in mortgage rates that is due to borrower risk, loan type, and aggregate time-series fluctuations.

The results in Table IX fail to support the view that more expensive mortgages are associated with better service quality. In fact, more expensive mortgages appear to be associated with more delays (though not statistically significant) and less satisfaction. Borrowers with a mortgage that is 100 bp more expensive are 13 percentage points less likely to report being very satisfied with their mortgage rate, and about 2 percentage points less likely to be very satisfied with their lender, the application process, and the closing

[48] One limitation of the NSMO data is that they do not contain a direct measure of points paid or received by the borrower. However, the controls for borrower wealth and expected time in the mortgage should help absorb differences in rates due to variation in points.

Table IX

Are Higher Mortgage Rates Associated with Better Service Quality?

Each row in this table represents a separate regression with the dependent variables listed in the first column. The displayed coefficients (β) are those on the borrower's mortgage rate, in percentage points. All outcomes are dummy variables, except "Overall satisfaction," which is the sum of the previous six satisfaction questions divided by six (i.e., its range is zero to one). The sample is restricted to first-lien 30-year fixed-rate mortgage loans for single-family principal residence properties, with no more than two borrowers, originated from 2013 through 2019 (the documentation question was not asked in every wave and thus has fewer observations). All regressions control for origination month fixed effects, survey wave fixed effects, county fixed effects, credit score (linear term plus dummies for 11 credit score bins), LTV (linear term plus dummies for six LTV bins), indicators for loan purpose (purchase, refinance, or cash-out refinance), nine loan amount categories, loan program (Freddie, Fannie, FHA, VA, FSA/RHS, other), first-time homebuyer status, single borrowers, log borrower income, self-employment status of borrower(s), respondent gender, race and ethnicity, whether the household owns four different types of financial assets, whether the household could pay their bills for three months without borrowing, metropolitan Community Reinvestment Act low-to-moderate income tract status, self-assessed creditworthiness, and the likelihood of moving, selling, or refinancing within a couple years. Data source: NSMO and the National Mortgage Database. Observations are weighted by NSMO sample weights. Robust standard errors are in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	β	SE	N	Outcome Mean
Did you experience delays in…				
…Your closing date	0.008	(0.008)	22,567	0.17
…Paperwork processing	0.009	(0.009)	22,567	0.23
Were you very satisfied with…				
…Your interest rate	−0.13***	(0.01)	22,567	0.69
…Your lender	−0.023***	(0.009)	22,567	0.77
…The application process	−0.019**	(0.01)	22,567	0.67
…The documentation process	−0.008	(0.011)	16,188	0.62
…The loan closing process	−0.019*	(0.01)	22,567	0.68
…The timeliness of disclosures	−0.007	(0.01)	22,567	0.68
Overall satisfaction	−0.035***	(0.009)	16,188	0.69

process. The other coefficients, though smaller in magnitude, also have a negative sign. Overall, rather than getting better service quality in return for a more expensive mortgage, these results point in the direction of borrowers getting worse service despite paying more.

D. Sophistication, Concentration, and Mortgage Rates

If consumers lack knowledge about mortgages or do not shop around effectively, this could lead to some degree of market power for lenders, even in low-concentration markets where many lenders offer mortgages. Furthermore, sophisticated borrowers may be better able to take advantage of more intense competition (lower market concentration). In this section, we examine these hypotheses using the NSMO data.

We start by studying the relationship between a borrower's mortgage contract rate and their shopping and knowledge. We estimate OLS regressions of

the form

$$Rate_{ijtw} = \beta X_i + \Gamma Z_{ij} + \alpha_t + \delta_w + \epsilon_{ijtw}, \tag{3}$$

where $Rate_{ijtw}$ is the contract rate on the mortgage for borrower i with loan characteristics j, loan origination month t, and responding to survey wave w, X_i are different measures of borrower i's shopping effort or knowledge about the mortgage market based on several questions in the NSMO, and Z_{ij} is a rich set of borrower and mortgage characteristics that could influence the pricing of the loan, similar to the set of controls described in equation (2) above (see the note to Table X for the full list of controls; recall that these are drawn mostly from loan-level administrative data, i.e., are not self-reported).

The first column of Table X shows how mortgage rates obtained by borrowers correlate with eight shopping and knowledge variables simultaneously. We think of the first four items as capturing shopping effort, while the other four capture knowledge about the mortgage market and their own mortgage. All are individually significant, suggesting that there are different dimensions to shopping and knowledge that can contribute to a borrower obtaining a low rate.[49] For instance, a borrower who is very familiar with market conditions may not need to consider more than one lender, if they can negotiate a good rate based purely on their knowledge.[50] Conversely, shopping alone does not guarantee a good rate if a borrower's knowledge is low (see also Malliaris et al. (2022)).[51]

In the second column, we regress $Rate_{ijctw}$ on a composite "sophistication index" measure, which we construct by taking the sum of six of the shopping and knowledge dummy variables, and then dividing by six so that the measure ranges from zero to one.[52] The coefficient on the sophistication index of -0.226

[49] In Internet Appendix Section VII, we provide a more complete description and summary statistics for each shopping and knowledge variable, as well as regression results for each of these variables individually. The coefficients are only slightly larger relative to the simultaneous regression in Table X. Borrowers may apply to 2+ lenders for reasons other than "better terms" (e.g., because they got turned down at one lender), which we condition on. We show that applying to 2+ lenders for other reasons is *positively* related to rates, in line with the findings of Agarwal et al. (2024).

[50] One may be concerned that borrowers who use a mortgage broker report only considering one lender even though the broker may be shopping across many lenders on the borrower's behalf. When we control for whether a borrower used a mortgage broker, our results remain virtually unchanged, and using a broker is associated with getting a slightly higher mortgage rate.

[51] In an earlier version of this paper (available at https://doi.org/10.17016/FEDS.2020.062), we provide a complementary analysis using data from the 2016 Survey of Consumer Finances. Consistent with the NSMO results, we find that borrowers who report shopping more, and borrowers with high financial literacy based on their answers to the Lusardi-Mitchell financial literacy questions, get significantly lower interest rates, even after controlling for loan characteristics, borrower credit risk, and borrower demographics.

[52] We include the dummy "considered 3+ lenders" but not the dummy for considering exactly two lenders; we also exclude the dummy for whether "most lenders offer the same rate" since this question was not asked in early waves. The median borrower has a sum of three (i.e., an index value of 0.5); about 14% of borrowers have a sum of one or zero, and about 8% of borrowers have a sum of five or six.

Table X

Sophistication, Lender Concentration, and Mortgage Rates

This table presents results of regressions in which the dependent variable is the borrower's mortgage interest rate, in percentage points. *Sophistication Index* is a composite of shopping/knowledge questions and ranges from zero to one (see text for details). *County HHI* is market concentration at the county-year level based on lender loan amounts, as of the year prior to origination, standardized to have a mean of zero and a standard deviation of one. The other independent variables are binary variables. The sample is restricted to first-lien 30-year fixed-rate mortgage loans for single-family principal residence properties, with no more than two borrowers, originated from 2013 through 2019 (waves 1 to 24; "most lenders offer same rate" was not asked in the first six waves of the survey, so we include a separate dummy for these missing observations). All regressions control for origination month fixed effects, survey wave fixed effects, county fixed effects, credit score (linear term plus dummies for 11 credit score bins), LTV (linear term plus dummies for six LTV bins), indicators for loan purpose (purchase, refinance, or cash-out refinance), nine loan amount categories, loan program (Freddie, Fannie, FHA, VA, FSA/RHS, other), first-time homebuyer status, single borrowers, log borrower income, self-employment status of borrower(s), respondent gender, race and ethnicity, whether the household owns four different types of financial assets, whether the household could pay their bills for three months without borrowing, Community Reinvestment Act low-to-moderate income tract status, self-assessed creditworthiness, applying to or more lenders for reasons other than "better loan terms," and the likelihood of moving, selling, or refinancing within a couple years. Observations are weighted by NSMO sample weights. Data Source: NSMO and the National Mortgage Database. Robust standard errors are in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	(1)	(2)	(3)
Considered 2 lenders	−0.022**		
	(0.009)		
Considered 3+ lenders	−0.044***		
	(0.012)		
Applied to 2+ lenders for better loan terms	−0.075***		
	(0.019)		
Used web/broker/friends to get info? A lot	−0.021***		
	(0.008)		
Familiar with mortgage rates? Very	−0.042***		
	(0.015)		
Most lenders offer same rate? No	−0.022**		
	(0.010)		
Knows their interest rate	−0.060***		
	(0.007)		
Answered whether rate is fixed or variable	−0.056***		
	(0.019)		
Sophistication Index		−0.226***	−0.219***
		(0.019)	(0.019)
Sophistication Index × County HHI (last year)			0.050**
			(0.020)
County HHI (last year)			−0.025*
			(0.014)
Adj. R^2	0.53	0.53	0.53
Observations	22,567	22,567	22,563

implies that the difference in rates between the most and least sophisticated borrowers is nearly 23 bp. This result suggests that variation across consumers in shopping and knowledge is an important contributor to mortgage rate dispersion, especially considering that our sophistication index is based on coarse responses to qualitative survey questions, likely leading to individual-specific noise and attenuation of the resulting coefficients.[53]

In column (3) of Table X, we interact the sophistication index with market concentration, measured as the county-level HHI of lender concentration in the year before origination and standardized to have a mean zero and standard deviation of one. Bear in mind that most of the U.S. population lives in counties with a low HHI, limiting the variation and range of HHI over which we can estimate the role of concentration. Nonetheless, we estimate a statistically significant and positive coefficient on the interaction term $HHI \times Sophistication$. This means that sophisticated borrowers tend to pay lower rates in less concentrated markets (HHI low) than in more concentrated markets (HHI high), while for less sophisticated borrowers the differences across market types are smaller. Thus, this finding is consistent with the idea that a reduction in market concentration may be less effective in promoting competition and limiting markups when borrowers are less sophisticated.

Further support for this story is provided in Internet Appendix Section VIII. We use the Optimal Blue locks data to assess whether the propensity of borrowers to get "good deals" (a rate residual less than −20 bp) or "bad deals" (a rate residual larger than +20 bp) varies with county HHI. We find that with higher market concentration, the likelihood of getting a good deal is significantly lower (while the average rate is not significantly related with HHI).

VI. Conclusion and Policy Implications

Our empirical results provide evidence that many borrowers from the most vulnerable part of the borrower population in the United States appear to overpay for mortgages—those that are most likely to be relatively low income, low net worth, and more likely to be first-time homebuyers. These are precisely the borrowers who various government programs attempt to subsidize. If they were to obtain mortgages from the lower end of the offer distribution, this would make their mortgage payments more affordable and leave them with more disposable income. Alternatively, the FHA and the GSEs could afford to raise their guarantee fees substantially without affecting final cost to borrowers if they were to choose cheaper lenders.

Given our findings, future research should consider the effects of policies that would help borrowers search and negotiate more effectively. This could

[53] For instance, respondents likely differ in what they view as using an information source "a lot" versus "a little," or being "very" versus "somewhat" familiar with a topic. In Internet Appendix Section VII.A, we provide evidence that mortgage knowledge and shopping are higher for jumbo borrowers relative to FHA borrowers, and for high-FICO borrowers relative to low-FICO borrowers, consistent with our earlier results that overpayment is correlated with loan program and credit score.

take the form of required information disclosure to borrowers of the rates available to them across different lenders in the same market (for instance, at the time they lock their rate). We recognize that this is not a straightforward endeavor given the multidimensional nature of mortgage pricing in the United States, but advances in technology may make this more feasible than in the past. Alternatively, future research could study whether the problem can be alleviated if the guaranteeing agencies were to impose requirements on the maximum locked-offer rate spreads they allow for loans to be securitized. Of course, to understand the effectiveness of such policies, one would need to consider general equilibrium effects on the offers that lenders make (as in Agarwal et al. (2024), Alexandrov and Koulayev (2017), or Guiso et al. (2022)).

The negative relationship between overpayment and the level of market rates that we document in Section III.B also matters for monetary policy transmission. Our findings imply that as rates fall (e.g., in response to central bank actions), borrowers tend to do worse relative to the rates available in the market, likely due at least in part to less shopping or negotiation. It follows that the contract rates they end up with do not fall as much as they could, based on lenders' offers, adding another friction to the pass-through of expansive monetary policy to the mortgage market.[54] On the other hand, the pass-through of increases in policy rates to rates on new mortgages may be dampened by more intense borrower shopping. This could be good or bad news for monetary policymakers, depending on whether slowing the housing market through higher mortgage rates is seen as desirable in a given situation or not.

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REFERENCES

Agarwal, Sumit, Itzhak Ben-David, and Vincent Yao, 2017, Systematic mistakes in the mortgage market and lack of financial sophistication, *Journal of Financial Economics* 123, 42–58.

Agarwal, Sumit, John Grigsby, Ali Hortaçsu, Gregor Matvos, Amit Seru, and Vincent Yao, 2024, Searching for approval, *Econometrica* 92, 1195–1231.

Agarwal, Sumit, Richard J. Rosen, and Vincent Yao, 2015, Why do borrowers make mortgage refinancing mistakes? *Management Science* 62, 3494–3509.

Alexandrov, Alexei, and Sergei Koulayev, 2017, No shopping in the U.S. mortgage market: Direct and strategic effects of providing information, Working paper, CFPB.

Allen, Jason, Robert Clark, and Jean-François Houde, 2014a, The effect of mergers in search markets: Evidence from the Canadian mortgage industry, *American Economic Review* 104, 3365–3396.

Allen, Jason, Robert Clark, and Jean-Francois Houde, 2014b, Price dispersion in mortgage markets, *Journal of Industrial Economics* 62, 377–416.

Ambokar, Sumedh, and Kian Samaee, 2019, Mortgage search heterogeneity, statistical discrimination and monetary policy transmission to consumption, Working paper, University of Pennsylvania.

Amel, Dean, Elliot Anenberg, and Rebecca Jorgensen, 2018, On the geographic scope of retail mortgage markets, FEDS Notes, Board of Governors of the Federal Reserve System.

[54] See Amromin et al. (2020) for a review of related work.

Amromin, Gene, Neil Bhutta, and Benjamin J. Keys, 2020, Refinancing, monetary policy, and the credit cycle, *Annual Review of Financial Economics* 12, 67–93.

Andersen, Steffen, John Y. Campbell, Kasper Meisner Nielsen, and Tarun Ramadorai, 2020, Sources of inaction in household finance: Evidence from the Danish mortgage market, *American Economic Review* 110, 3184–3230.

Argyle, Bronson, Taylor Nadauld, and Christopher Palmer, 2022, Real effects of search frictions in consumer credit markets, *Review of Financial Studies* 36, 2685–2720.

Armstrong, Mark, and John Vickers, 2022, Patterns of competitive interaction, *Econometrica* 90, 153–191.

Baye, Michael R., John Morgan, and Patrick Scholten, 2006, Information, search, and price dispersion, in T. Hendershott, ed.: *Handbook of Economics and Information Systems* (Elsevier).

Bhutta, Neil, and Aurel Hizmo, 2021, Do minorities pay more for mortgages? *Review of Financial Studies* 34, 763–789.

Bhutta, Neil, Steven Laufer, and Daniel Ringo, 2017, Residential mortgage lending in 2016: Evidence from the Home Mortgage Disclosure Act data, *Federal Reserve Bulletin* 103, 1–27.

Bhutta, Neil, and Daniel Ringo, 2021, The effect of interest rates on home buying: Evidence from a shock to mortgage insurance premiums, *Journal of Monetary Economics* 118, 195–211.

Buchak, Greg, Gregor Matvos, Tomasz Piskorski, and Amit Seru, 2018, Fintech, regulatory arbitrage, and the rise of shadow banks, *Journal of Financial Economics* 130, 453–483.

Carlson, John A., and R. Preston McAfee, 1983, Discrete equilibrium price dispersion, *Journal of Political Economy* 91, 480–493.

CFPB, 2020, Data point: 2019 mortgage market activity and trends, Technical report, Consumer Financial Protection Bureau.

Choi, James J., David Laibson, and Brigitte C. Madrian, 2010, Why does the law of one price fail? An experiment on index mutual funds, *Review of Financial Studies* 23, 1405–1432.

Coen, Jamie, Anil Kashyap, and May Rostom, 2023, Price discrimination and mortgage choice, National Bureau of Economic Research Working Paper 31652.

Damen, Sven, and Erik Buyst, 2017, Mortgage shoppers: How much do they save? *Real Estate Economics* 45, 898–929.

Drechsler, Itamar, Alexi Savov, and Philipp Schnabl, 2017, The deposits channel of monetary policy, *Quarterly Journal of Economics* 132, 1819–1876.

Fuster, Andreas, Laurie Goodman, David Lucca, Laurel Madar, Linsey Molloy, and Paul Willen, 2013, The rising gap between primary and secondary mortgage rates, *Federal Reserve Bank of New York Economic Policy Review* 19, 17–39.

Fuster, Andreas, Stephanie H. Lo, and Paul S. Willen, 2024, The time-varying price of financial intermediation in the mortgage market, *Journal of Finance* 79, 2553–2602.

Gabaix, Xavier, David Laibson, Deyuan Li, Hongyi Li, Sidney Resnick, and Casper G. de Vries, 2016, The impact of competition on prices with numerous firms, *Journal of Economic Theory* 165, 1–24.

Gomes, Francisco, Michael Haliassos, and Tarun Ramadorai, 2021, Household finance, *Journal of Economic Literature* 59, 919–1000.

Goodman, Laurie S., 2017, Quantifying the tightness of mortgage credit and assessing policy actions, *Boston College Journal of Law & Social Justice* 37, 235–265.

Guiso, Luigi, Andrea Pozzi, Anton Tsoy, Leonardo Gambacorta, and Paolo Emilio Mistrulli, 2022, The cost of steering in financial markets: Evidence from the mortgage market, *Journal of Financial Economics* 143, 1209–1226.

Gurun, Umit G., Gregor Matvos, and Amit Seru, 2016, Advertising expensive mortgages, *Journal of Finance* 71, 2371–2416.

Hastings, Justine S., Brigitte C. Madrian, and William L. Skimmyhorn, 2013, Financial literacy, financial education, and economic outcomes, *Annual Review of Economics* 5, 347–373.

Hortaçsu, Ali, and Chad Syverson, 2004, Product differentiation, search costs, and competition in the mutual fund industry: A case study of S&P 500 index funds, *Quarterly Journal of Economics* 119, 403–456.

Iscenko, Zanna, 2018, Choices of dominated mortgage products by UK consumers, Occasional Paper 33, Financial Conduct Authority.

Jiang, Erica, Gregor Matvos, Tomasz Piskorski, and Amit Seru, 2020, Banking without deposits: Evidence from shadow bank call reports, National Bureau of Economic Research Working Paper 26903.

Keys, Benjamin, Devin Pope, and Jaren Pope, 2016, Failure to refinance, *Journal of Financial Economics* 122, 482–499.

Liu, Lu, 2019, Non-salient fees in the mortgage market, Working paper, Imperial College London.

Malliaris, Steven, Daniel A. Rettl, and Ruchi Singh, 2022, Is competition a cure for confusion? Evidence from the residential mortgage market, *Real Estate Economics* 50, 206–246.

McManus, Doug, Liyi Liu, and Mingzhe Yi, 2018, Why are consumers leaving money on the table? Freddie Mac Economic & Housing Research Insight.

Stango, Victor, and Jonathan Zinman, 2016, Borrowing high versus borrowing higher: Price dispersion and shopping behavior in the U.S. credit card market, *Review of Financial Studies* 29, 979–1006.

Syverson, Chad, 2019, Macroeconomics and market power: Context, implications, and open questions, *Journal of Economic Perspectives* 33, 23–43.

Woodward, Susan E., and Robert E. Hall, 2012, Diagnosing consumer confusion and sub-optimal shopping effort: Theory and mortgage-market evidence, *American Economic Review* 102, 3249–3276.

Wright, Randall, Philipp Kircher, Benoît Julien, and Veronica Guerrieri, 2021, Directed search and competitive search equilibrium: A guided tour, *Journal of Economic Literature* 59, 90–148.

Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Appendix S1: Internet Appendix.
Replication Code.